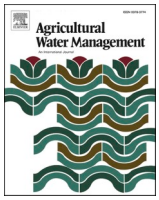
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Spatio-temporal estimation of green and blue water consumptions and water and land productivity using satellite remote sensing datasets and WA+ framework: A case study of the Mahi Basin, India

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ABSTRACT

The agricultural activities contribute to the largest share of water consumption in the arid and semi-arid basins. In this study, we demonstrate the application of Water Accounting Plus (WA+) for estimation of the green water consumption (ETGreen) and blue water consumption (ETBlue) for assessing the water productivity (WP) and land productivity (LP) to identify the bright-spots and hot-spots at the district administrative unit level for effectively managing the scarce water resources and sustaining food security in a highly non-resilient semi-arid basin of India. The WA+ framework uses satellite remote sensing datasets from different sources for this purpose and we used the data from 2003 to 2020. The long-term average of ETGreen and ETBlue in the Mahi basin is found to be 15.8 km^3/year and 12.32 km^3/year, respectively. The blue water index (BWI) and green water index (GWI) in the basin vary from 0.282 to 0.598 and 0.40–0.72. We found that the BWI is highest for the districts of Gujarat, whereas, the GWI is highest for the districts of Madhya Pradesh. The long-term average of the LP and WP for both the irrigated and rainfed cereals in the basin is found as 2287.71 kg/ha & 1713.62 kg/ha and 0.721 kg/m^3 & 0.483 kg/m^3 , respectively from 2003 to 2020. The WP (rainfed) of all the districts of the Gujarat is comparatively lower (varying from 0.34 kg/m^3 to 0.5 kg/m^3) than the districts of the Madhya Pradesh (varying from 0.59 kg/m3 to 0.70 kg/m^3) and the Rajasthan (varying from 0.48 kg/m^3 to 0.73 kg/m^3). Based on the results, we found that the Ratlam district of the Madhya Pradesh has both highest LP and WP (irrigated) as 2573.96 kg/ha and 2.14 kg/m^3, respectively among all the districts of the Mahi basin, and hence it is classified as the 'Bright spot-district'. The Anand district is found to have the lowest WP and LP as 0.44 kg/m^3 and 2467.51 kg/ha, respectively and hence it is classified as the 'hot spot-district'. For rainfed cereals, we found that the Neemuch district of Madhya Pradesh has the highest WP and LP as 0.59 kg/m^3 and 1948.13 kg /ha, respectively, and the Anand district with the lowest WP as 0.34 kg/m^3 and LP of 1572.21 kg/ha, respectively. Therefore, we classified the Neemach district as the 'Bright spot-district' and the Anand district as the hot spot- district for rainfed cereals. These findings will help develop sustainable and actionable agricultural water management plans by the policymakers and stakeholders in the basin.

1. Introduction

Increasing water productivity (WP) and land productivities (LP) for sustaining food security and freshwater availability is a significant challenge to the current and future generations (Safi et al., 2022). The water availability for agricultural production will decrease due to increased competition from industrial and domestic sectors (Zwart and Bastiaanssen, 2007) and this will eventually put excess pressure on the farmers. This is the reason that there is a growing emphasis on increasing WP in agriculture (Kang et al., 2017; Marston et al., 2020; Molden, 2007; Bastiaanssen et al., 2003). WP is represented as the ratio of the crop yield to the green and blue water consumption (i.e., ETgreen and ETBlue), where ETGreen and ETBlue represent the actual evapotranspiration (ET) due to green water and blue water, respectively

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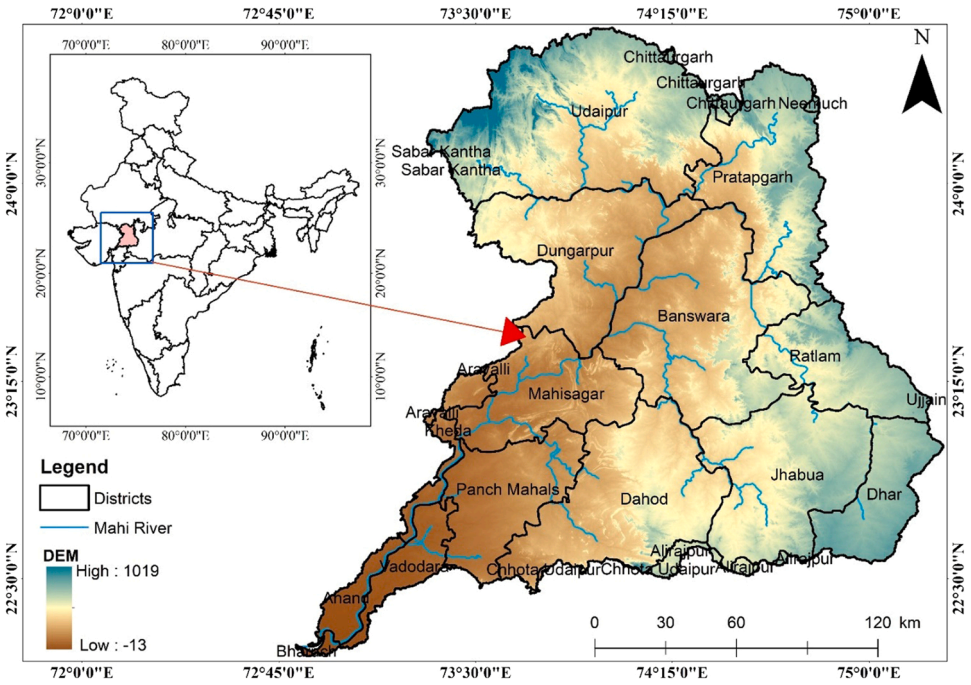


Fig. 1. Location of the Mahi River Basin.

(Falkenmark and Rockström, 2006; Velpuri and Senay, 2017; Mekonnen and Hoekstra, 2011; 2012; Chukalla et al., 2015; Mekonnen and Hoekstra, 2011; Msigwa et al., 2021; Singh et al., 2021). WP measures the capability of a system to convert water consumed into goods and services (Molden, 1997). WP also helps establish a linkage between agricultural water management with water security, food security, and economic development (Bastiaanssen et al., 2003). Improving WP will help to reduce the need for additional resources in both irrigated (blue water) and rainfed (green water) systems (Molden et al., 2010) and its benchmarking will help identify the productivity levels for improving the sustainability of the water use in agriculture (Marston et al., 2020).

Worth mentioning José Graziano da Silva, Former Director-General, Food and Agriculture Organization (FAO), and as quoted in Kang et al. (2017) that "There is no food security without water security". To the fact that agriculture consumes about 70% of the total water use worldwide, and therefore, improving agricultural water productivity is an important measure for ensuring global water safety and food security (Kang et al., 2017). The Government of India (Economic Survey, 2018–19) also emphasized shifting from LP to blue WP (i.e., irrigation WP). However, at the same time, improvements in the green WP (i.e., rainfed WP) are also very much helpful in the management of green water, especially for arid areas (Rockström and Barron, 2007). An increase in WP will help to generate net water savings, which can be used further for bringing more area under irrigation, diverting it for other purposes (domestic, industrial, and environmental), and reducing pressure in regions with higher water stress or scarcity (Singh et al., 2021; Zheng et al., 2020; Fader et al., 2011). The improvements in WP can improve production and profitability (Clothier et al., 2020). Therefore, it necessitates exploring the advanced tools and techniques for quantifying WP and LP in agriculture for solving future water and food challenges. However, the major challenge in quantifying WP and LP (spatially and temporally) is the availability of accurate and extensive data on yield and agricultural water use (Kelley and Olson, 2022). However, this challenge is being addressed to the largest extent by making use of the freely available (most of) remote sensing and earth observation datasets by many agencies such as National Aeronautics and Space Administration (NASA), European Space Agency (ESA), and other space agencies and academic institutions worldwide. These earth observation datasets are available in the near real-time at varying spatial and temporal resolutions (Al-Bakri et al., 2022; Chawla et al., 2020; Foster et al., 2020; Karimi et al., 2013; Zhang et al., 2022).

Long back, International Water Management Institute (IWMI) developed water accounting (WA) procedure (Molden, 1997) to identify water consumption patterns, WP & LP, flows, and fluxes (Delavar et al., 2020). However, one of the major limitations of the WA procedure was the availability of the observed datasets to derive the required outputs at the desired resolutions (Karimi et al., 2013a,b). To rectify this limitation, WA procedure was later improved by IMWI, IHE Delft, FAO by making use of satellite remote sensing datasets and open access datasets, and the procedure was re-named as the water Accounting Plus (WA+) framework. Therefore, keeping in view of the aforementioned discussion, in this study WA+ framework has been applied to the Mahi river basin, a semi-arid basin in the western part of India, with the following objectives as:(i) to study the land-use impact over the water consumption patterns based on water-management classes (ii) to partition ET fluxes into evaporation (E), transpiration (T) and interception (I) fluxes, (iii) to estimate ETGreen, ETBlue and their spatial variability in the basin, and finally (iv) to assess the LP and WP of the major crop (mainly cereals) and their spatial and temporal variability in the basin.

The organizational structure of the paper is as follows. Section 1 highlights the importance of WP and LP to address the issues of food security and freshwater availability. This section also highlights the usefulness of the remote sensing earth observation datasets in the estimation of spatial and temporal variability of the WP and LP in the basin. Section 2 deals with the study area. Section 3 gives a detailed description of the WA+ framework describing all the worksheets, their input data requirements, and outputs. Section 4 deals with the methodology used in this study. This section gives the basic equations used for calculating the water consumption pathways (i.e., E, T & I) and LP and WP in the basin and different performance indicators. This section also deals with the estimation of the ETGreen and ETBlue using Budyko hypothesis. Section 5 describes various database and inputs used in the study. Section 6 discusses the important results obtained from this study and their implications in agricultural water management and enhancing WP and LP for sustaining water resources and ensuring food security in the basin. Section 7 highlights the important conclusions drawn from the study

Table 1
District-wise long-term mean rainfall for the period of 2003–2020.

Districts	Mean rainfall (mm)
Anand	100.12
Aravalli	83.21
Bharuch	101.15
Chhota Udaipur	96.58
Dahod	87.00
Kheda	88.17
Mahisagar	86.54
Panch Mahals	90.76
Sabar Kantha	81.25
Vadodara	95.20
Alirajpur	86.73
Dhar	85.60
Jhabua	87.35
Neemuch	82.51
Ratlam	87.81
Banswara	90.14
Chittaurgarh	78.94
Dungarpur	79.28
Pratapgarh	83.59
Udaipur	74.62

and put forward relevant suggestions. Finally, Section 8 deals with the limitations and the potential for future research.

2. Study area

The Mahi basin situated in the western part of India, falls under the semi-arid region (Fig. 1). This basin is an inter-state basin, of which 42 % and 40 % of the area falls in the states of Gujarat and Rajasthan, respectively, and 18 % area in the state of Madhya Pradesh. The Mahi basin has a catchment area of 33,839.22 km^2 and lies between 72°21′E to 75°19′ E and 21°46′ N to 24°30′ N. The upper half of the basin in Madhya Pradesh state comprises undulating land with ridges and valleys, whereas the lower half is a plain area. A major part of the basin area lies in the elevation zones of 100–500 m. The annual average rainfall & ET in the basin varies from 800 to 1300 mm and 700–900 mm, respectively. Table 1 shows the district-wise average rainfall from 2003 to 2020. ET is about 65–80 % of the total rainfall. A major part of the basin receives rainfall during July to September. The mean annual temperature of the basin varies from 26 °C to 27 °C. The Mahi basin is agriculture dominated with land use land cover (LULC) of the basin comprises of 65 % of agricultural land, 11 % forest cover & about 2.9 % water bodies. Major crops of the basin include cereals (Rice, maize, wheat). The leaf area index (LAI) ranges from 0.33 to 1.5 m^2/m^2, which is an indicator of the low canopy cover in the basin. The Mahi basin falls under the non-resilient ecosystem (Sharma et al., 2017), and because of this, it faces seasonal droughts regularly, and some of the areas come under barren land and undulating areas. Hence, there is a need for thorough investigations on water uses in agriculture and quantification of WP and LP (for irrigated and rainfed agriculture both) and basin performance indicators that can help the basin authorities and other stakeholders to develop actionable plans for ensuring food security and agricultural water management.

3. Water accounting plus (WA+) framework

WA+ framework is developed by IHE in collaboration with IWMI, FAO, World Water Assessment Programme (WWAP) (Karimi et al., 2013). WA+ framework is useful for the systematic assessment of water consumption patterns, WP & LP, flows, and fluxes (Delavar et al., 2020) and shares the structural. Recently, the WA+ framework has emerged as a key tool for understanding the water-food nexus and identifying the gaps and inefficiencies in water uses. WA+ framework is based on the fundamental concept of water 'depletions' as that of IWMI-WA rather than 'withdrawals' and segregates these depletions into ETgreen and ETBlue using the popular Budyko (1974) hypothesis. Notably, the water depletions in agriculture can be represented by the measurable ET process through satellite remote sensing (Karimi et al., 2013a,b) WA+ establishes a linkage between the landuse classes and water depletions and distinguishes between irrigated and rainfed agriculture using the blue and green water concept. Due to its ability to employ satellite-driven information, WA+ is especially appropriate and helpful for applications at large scales and in data-scarce regions (Delavar et al., 2020).

WA+ framework provides seven key fact sheets (Delavar et al., 2020; Salvadore et al., 2020) as: (i) resources-base, (ii) evapotranspiration, (iii) agricultural services or productivity, (iv) withdrawals, (v) surface water, (vi) groundwater and (vii) ecosystem services. The resource base sheet gives us the overall water resources scenario of the basin in terms of the unmanageable, manageable, exploitable, reserved, utilized, and utilizable flows. This sheet uses precipitation, ET, PET, change in storage, outflows and net withdrawals, and many other datasets as given in Table A1. The evapotranspiration is the dominant eco-hydrologic process in any basin. The 'evapotranspiration sheet' provides information on the water consumption patterns in the basin as per the four major land and water uses classes (sometimes also referred as water management classes, WMCs), i.e., protected landuse (PLU), utilized landuse (ULU), modified landuse (MLU), and managed water use (MWU)). A brief description of these four major landuse classes (also known as water management clusters) will be given at the end of this section. This sheet also gives which fraction of ET is manageable (i.e. ET from ULU), managed (i.e., ET from MLU and MWU) and non-manageable (i.e., ET from PLU s). The partitioning of ET into beneficial (Transpiration: T) and non-beneficial consumption (soil and water evaporation: E and canopy interception: I) is the unique feature of this sheet. The ET partitioning into the soil and water evaporation (E), canopy interception evaporation (I), and transpiration (T) yield both comprehensive insight into hydrological processes and better water management (Nguyen et al., 2021).

The 'Agricultural Services or Productivity sheet' provides information on the spatial and temporal variability of WP and LP and help identify the hot spots and bright spots which will help manage improved water management action plans (engineering, agronomical and societal) for sustaining water resources and food security in the basin. This sheet also gives the spatial and temporal estimates of WP and LP individually from rainfed and irrigated agriculture hence providing scope for devising plans (individually as well as jointly) for agricultural water management. The LP and WP are calculated based on the representative crop as cereals, although different crops are also cultivated in this region, rice and wheat are considered as the major crop. The withdrawals sheet (also known as the Utilized flow sheet) summarizes all the man-made withdrawals (surface and groundwater), consumed and non-consumed water, and recoverable and non-recoverable water. The surface water sheet provides information on the surface water availability (natural) at the basin and sub-basin levels as per the four WMCs, i.e., PLU, ULU, MLU, and MWU. The groundwater sheet summarizes the groundwater flows and storage within the basin and facts on the vertical recharge and groundwater withdrawals. The ecosystem services sheet provides information on ecosystem services produced within a basin by consuming water resources. However, this fact sheet is still in the developmental phase at IHE Delft. The inputs and outputs of all the above key fact sheets are given in Table A1. Appendix A also summarizes the definitions of the different terms used in the WA+ framework.

As discussed above, the WA+ framework has four major land and water uses classes (WMCs) as: (i) PLU: represents areas such as National Parks & Wildlife Sanctuaries that allows minimal disturbances by human, (ii) ULU: represents areas such as natural pastures, forests which allows little interference by humans and provides services such as food, fodder, and fire woods, (iii) MLU: In this class, the hydrological processes are significantly modified by human activity and the rainfed cropping system is the typical examples of this group and finally (iv)

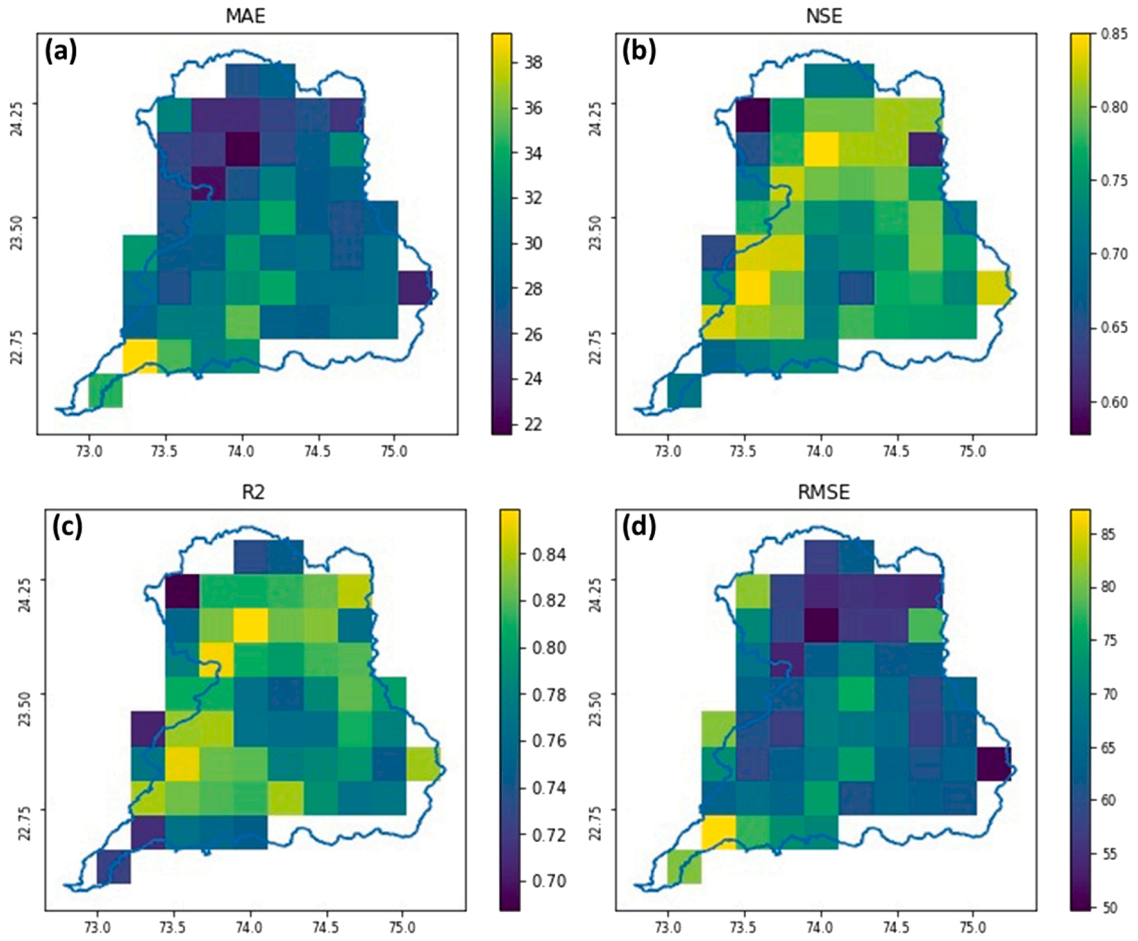


Fig. 2. Comparison of CHIRPS with IMD using (a) MAE, (b) NSE, (c) R2, and (d) RMSE.

MWU: In this class, the natural water cycle is changed thoroughly by the irrigation systems and water infrastructural activities. Irrigated agriculture is the major landuse class of this class. All these four major classes are further divided into 80 LULC classes. The details on these 80 LULC classes can be taken from Karimi et al. (2013a), Michailovsky et al. (2020) and Mul et al. (2020).

4. Materials and methodology

In this study, we are focusing on two key fact sheets, i.e., the evapotranspiration sheet and the agricultural services or productivity sheet. The evapotranspiration sheet is used for estimating water consumption patterns and partitioning of ET into E, T, and I flux, The Budyko hypothesis and the agricultural services sheet are used for the estimation of ETGreen and ETBlue for assessing the spatial and temporal variability of WP and LP and identification of gaps for sustaining water resources and food security in the Mahi basin. This analysis will help identify the WP gaps and associated LP gaps in the basin and show up to what extent the water in the agricultural sector is used productively. This section discusses the datasets and the methodology used to address the objectives of this study as mentioned in the beginning of this section. Fig. 5 shows the Flowchart of the WA+ Framework used in this study.

4.1. Datasets used

The following datasets are used in this study from 2003 to 2020 as: (i) Precipitation (P) from Climate Hazards Group Infrared Precipitation with Stations (CHIRPS), (ii) Actual evapotranspiration (ET) Simplified Surface Energy Balance ($SSEB_{OP}$) from USGS FEWS, (iii) Potential evapotranspiration (PET) extracted from Global Data Assimilation System (GDAS) analysis fields from NOAA, (iii) Leaf Area Index (LAI) from MODIS (MOD15), (iv) Net Primary production (NPP) and Gross primary production (GPP) from MODIS (MOD17). All the input datasets used in this study are in raster (GeoTIFF) format.

WA+ land use utilizes different land-use products to classify the land use classes as discussed in Section 3. These different landuse products maps are: (i) NRSC LULC (ii) IWMI irrigated crop map (iii) MIRCA Rainfed and irrigated crop maps (MIRCA 2000) (iv) WDPA (World Database on protected areas (v) World data on Permanent water bodies (perWB), JRC-EU, (vi) Indian population map (WorldPOP), (vii) ESA Globcover map. Table A2 (Appendix A) also shows the above-mentioned data description and sources in detail.

4.2. P, ET, LAI, dry matter (NDM) and mean water yield variability

Worth mentioning, precipitation is the key input in the WA+ as the other hydrological processes are dependent on it. In this study, we used CHIRPS precipitation datasets from 2003 to 2020. Before using the CHIRPS datasets, we evaluated it's applicability with India Meteorological Department (IMD) gridded rainfall datasets, and the goodness-of-fit is evaluated in terms of nash shutcliffe efficiency (NSE) (Eq.1), coefficient of determination (R2) (Eq.2), root mean square error (RMSE) (Eq.3), mean absolute error (MAE) (Eq.4) for Mahi river basin (Faridzad, 2018) as shown in Fig. 3. These statistical indices are expressed here as:

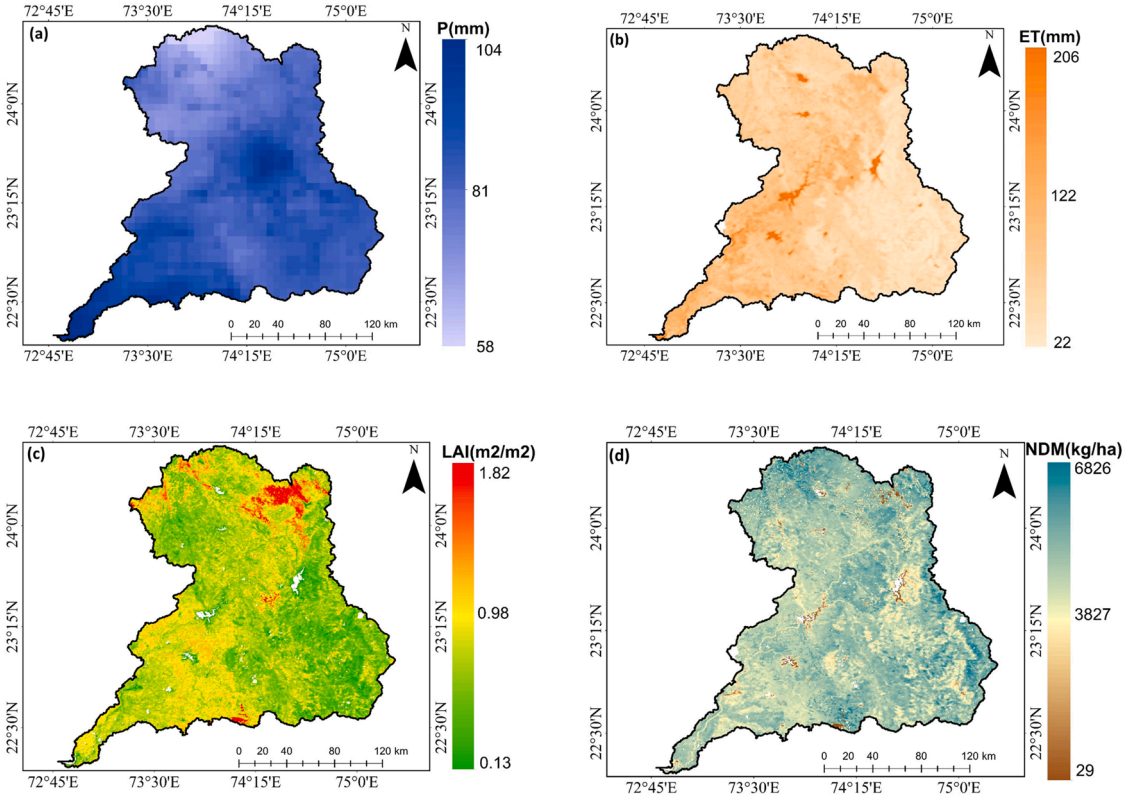


Fig. 3. Spatial variation in annual average (2003–2020): (a) P, (b) ET, (c) LAI, and (d) NDM.

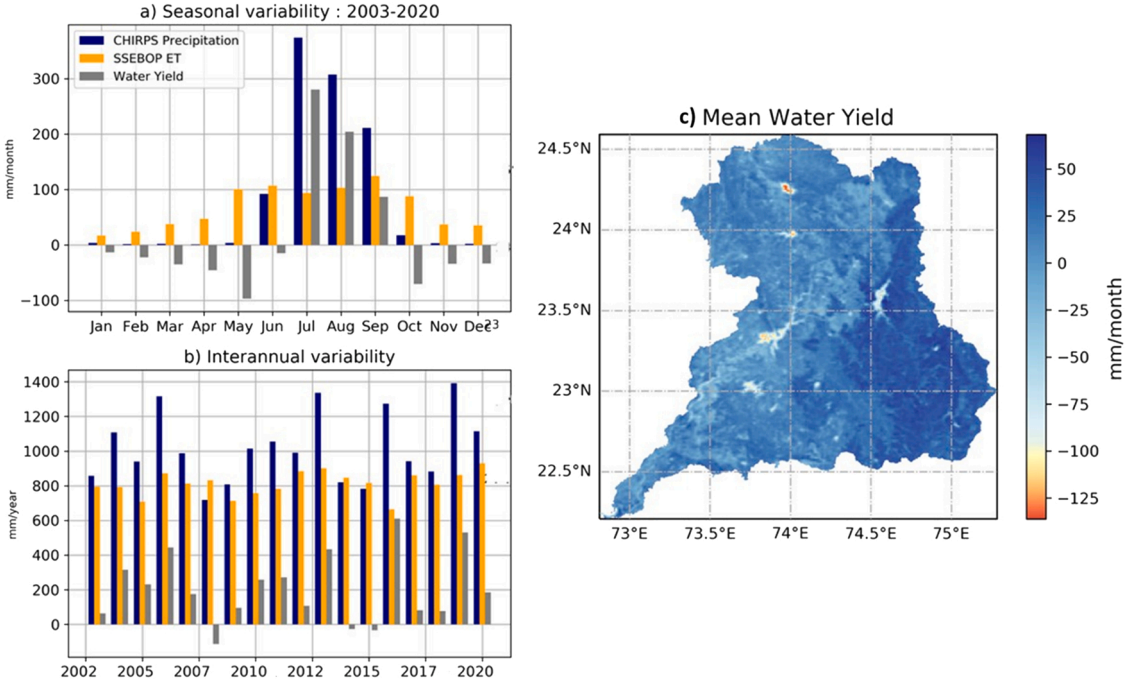


Fig. 4. (a&b): Seasonal and inter-annual variability of P, ET, and water yield, (c) Spatial variation in annual average in water yield for the period of 2003–2020.

$$NSE = 1 - \frac{\sum_{t=1}^{T}\left(P_{o(t)} - P_{m(t)}\right)^2}{\sum_{t=1}^{T}\left(P_{o(t)} - P_{o(avg)}\right)^2} \quad (1)$$

$$R^2 = \left(\frac{\sum\left[\left(P_{o(t)} - P_{o(avg)}\right) * \left(P_{m(t)} - P_{m(avg)}\right)\right]}{\sqrt{\sum\left[\left(P_{o(t)} - P_{o(avg)}\right)^2 * \left(P_{m(t)} - P_{m(avg)}\right)^2\right]}} \right)^2 \quad (2)$$

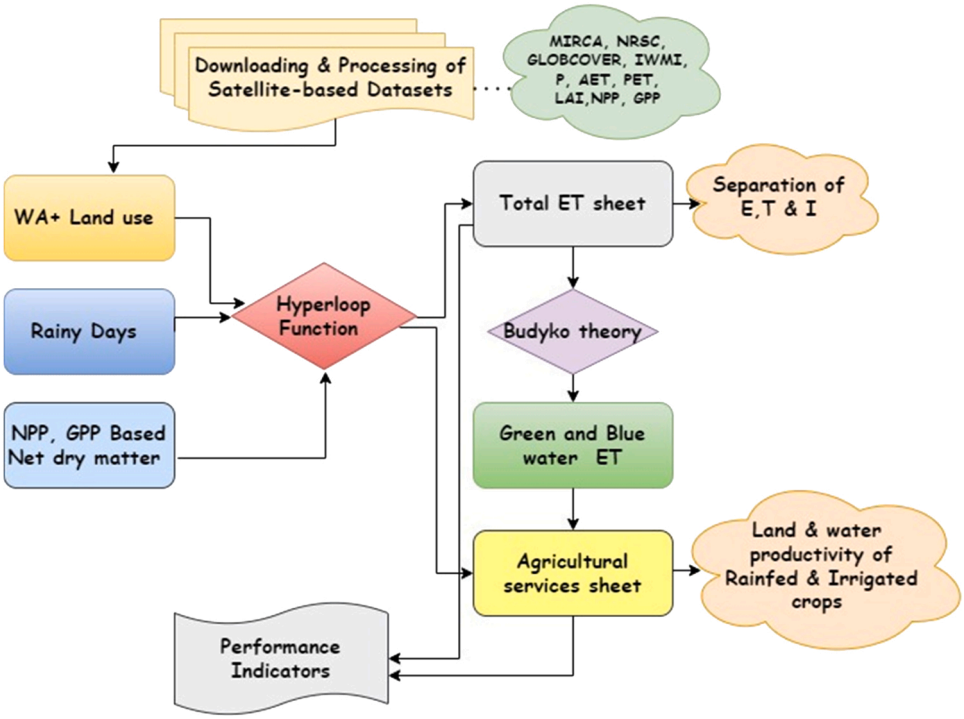


Fig. 5. Flowchart of WA+ framework methodology.

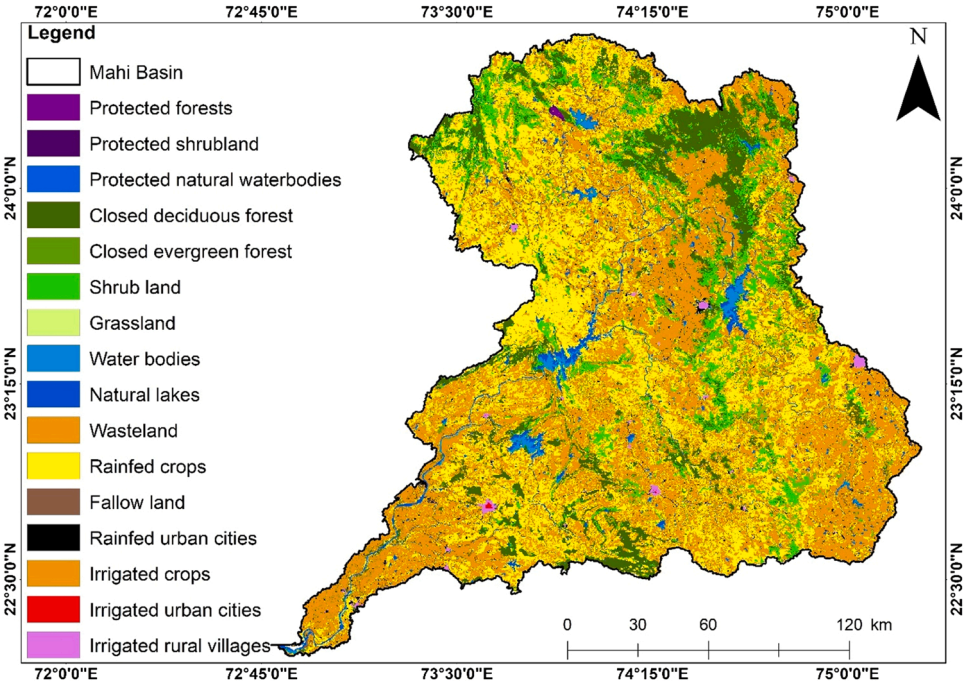


Fig. 6. WA+ land use land cover map.

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(P_e - P_i)^2}{n}} \quad (3)$$

$$MAE = \frac{\sum_{t=1}^{T} (P_{o(t)} - P_{m(t)})}{n} \quad (4)$$

where, Po(t), Pm(t), Po(avg) & Pm(avg) are the observed, average modelled (satellite) average rainfall in mm at the time steps (t), and n is the number of observations. Fig. 2 shows CHIRPS dataset has higher values of NSE and R^2 and lower values of MAE and RMSE with the IMD dataset. Prakash, (2019) and Gupta et al. (2019) also reported a similar performance of the CHIRPS dataset for India. Recently, Bhattacharyya et al., (2022) also found that the CHIRPS dataset represents better annual and seasonal rainfall characteristics across India than the other satellite and re-analysis precipitation datasets with respect to the IMD dataset. Therefore, we used CHIRPS datasets for WA+ applications in this study.

Table 2
Long-term average AET, T and I from the four major land and water use classes.

Land and water use classes (WMCs)	Land use classes	Area (km^2)	ET (km^3/year)	T (km^3/year)	I (km^3/year)
Protected land use (PLU)	Protected forests	12.19	0.0134	0.0082	0.001
	Protected shrub land	6.86	0.0069	0.0041	0.000
	Protected natural water bodies	1.45	0.0014	0.0010	0.000
Total		20.5	0.0217	0.0133	0.001
Utilized land use (ULU)	Closed deciduous forest	3504.56	3.0695	1.3719	0.128
	Closed evergreen forest	34.20	0.0391	0.0119	0.001
	Shrub land & mesquite	2912.06	2.0716	1.0151	0.091
	Meadows & open grassland	55.94	0.0475	0.0290	0.002
	Waste land	312.80	0.5227	0.0406	0.004
	Natural lakes	1093.04	1.1321	0.1915	0.024
	Moorland	3478.66	2.4763	1.0588	0.093
Total		11,391.26	9.3588	3.7188	0.343
Modified land use (MLU)	Rainfed crops - cereals	10,454.68	7.9595	3.6011	0.296
	Fallow & idle land	997.74	0.7380	0.2643	0.026
	Rainfed homesteads and gardens (rural villages) - outdoor	284.83	0.2484	0.0619	0.007
		11,737.25	8.9459	3.9273	0.329
Managed water use (MWU)	Irrigated crops - cereals	10,553.32	9.0487	3.6630	0.285
	Irrigated homesteads and gardens (urban cities) - outdoor	5.21	0.0051	0.0003	0.000
	Irrigated homesteads and gardens (rural villages) - outdoor	114.52	0.1007	0.0104	0.002
		10,673.05	9.1545	3.6737	0.287

The spatial variability of the P, ET, LAI, and net dry matter (NDM) is shown in Fig. 3(a-d). The NDM (net dry matter) is calculated using NPP and GPP datasets. The average annual rainfall in the basin is about 1060 mm/year. The northern part of the basin (the Rajasthan state) receives low rainfall as compared to the western (Gujarat) and eastern (Madhya Pradesh) parts of the basin. This spatial variability of rainfall is also affecting the biophysical parameters in the basin such as ET, LAI, and NDM as shown in Fig. 3(b-d). The average ET of the basin is 857 mm/year, which is about 80 % of the average annual rainfall. The south-western (Gujarat) part of the basin shows the highest ET followed by the parts of Rajasthan as compared to the Eastern parts of the basin (Madhya Pradesh). Annual variability of P, ET, and water yield (P-ET) for the Mahi basin is also shown in Fig. 4. For the years, when P is lower than ET, the water yield is negative in the Mahi basin.

4.3. Methodological steps

This section details the general steps followed to address the objectives of this study. The flow chart of these steps is also given in Fig. 5.

4.3.1. Preprocessing of open-source satellite datasets

All the satellite and open-access datasets were downloaded and processed to the same resolution of 250 m x 250 m for further application in the study.

4.3.2. Estimation of NDM

The NDM (kg/ha) (on monthly scale) is derived using NPP (kgC/m^2) (a MODIS product available on annual scale) and GPP (kgC/m^2) (a MODIS product available on 8-daily product is converted on monthly scale). NPP describes the net amount of carbon (C) that is absorbed by vegetation through assimilation and respiration. A detailed description on this can be found in Peiser and Bastiaanssen (2015).

4.3.3. Generation of WA+ land use land cover map

As discussed in Section 3, WA+ framework has four main land and water uses classes (also known as WMCs) as PLU, ULU, MLU and MWU based on the degree of the human disturbances altering the natural hydrological cycle and associated processes.

The coupling between the landuse and land cover with the water consumption patterns is an essential information, particularly, for the arid regions evaluation of water management strategies and policy making (Peiser and Bastiaanssen, 2015). WA+ LULC induces space for different management options for different LULC classes. Further, it segregates the rainfed (green water) and irrigated agriculture (green water + blue water) which makes it unique as joint management of green water and blue water may decrease the blue water dependency thus redefining the integrated water resources management (IWRM) (Falkenmark and Rockström, 2006). The developed WA+ based LULC (WALU) map for the Mahi basin is shown in Fig. 6. The PLU, which consists of 3 subclasses (protected forests, protected shrublands, and protected waterbodies), accounts for just 0.06 % of total land use, whereas the ULU, which consists of 7 subclasses, accounts for 33.68 % of total land use on the basin. The forest cover amounts to 10.50 % of the total. The MLU, which is mostly comprised of rainfed crops, is about 34.70 % of the basin; the principal rainfed crop recognized in the basin are cereals, as determined by the MIRCA crop map. The MWU class includes only 3 classes, which are mostly irrigated crops and irrigated built-up areas, accounting for 31.56 % of the basin. Table 2 also shows the four landuse and water use classes and major crop acreage in the basin.

4.3.4. Partitioning of ET into E, T & I

An accurate ET partitioning is highly desirable for research related to agriculture, climate and land-use change, hydrology, and water resource availability (Talsma et al., 2018). ET partitioning is also a crucial factor in global climate models (Lawrence et al., 2007). ET partitioning can be used to separate the productive and un-productive (or beneficial or non-beneficial) fluxes (Delavar et al., 2020; Sutanto et al., 2012). This partitioning is also very essential to improve WP in agriculture, especially in case of water scarcity, knowledge about water fluxes in the vadose zone is essential. Kool et al. (2014) reviewed 52 ET partitioning publications and stated that ET partitioning is important to assess biomass production and the allocation of increasingly scarce water resources.

WA+ enables us to identify the water consumption patterns and provides better insights for water resource management. Based on the WA+ based LULC classification, the 'evapotranspiration' sheet depicts water depletion and specifies components of water utilization that can be managed, manageable, and non-manageable. It divides ET into E, T and I, and then distinguishes between beneficial and non-beneficial water consumption. For instance, E is taken as non-beneficial component of ET, whereas, T is taken as the beneficial component of ET contributing to food production, and the economy. The estimation of I depends on the LAI, P, and the number of rainy days and is estimated by the relationship given by Von Hoyningen-Hüne (1983) and Braden (1985) as:

$$Im = \left(1 - \frac{1}{1 + \frac{Pm}{Nm} * \left(1 - e^{(-0.5 * LAIm)}\right) * \frac{1}{LAIm}}\right) * LAIm * Nm \quad (5)$$

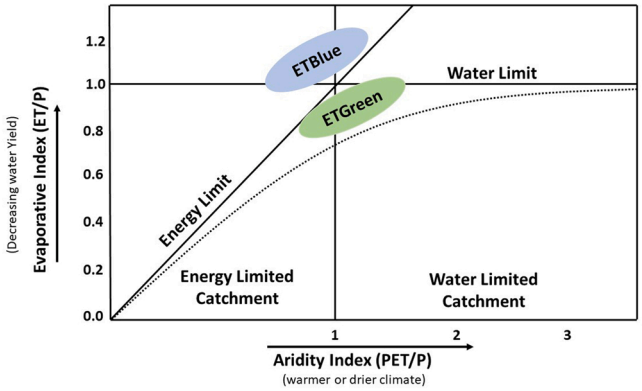


Fig. 7. Budyko curve for the estimation of ETGreen and ETBlue.

where Im represents monthly interception (mm), Pm & Nm denotes the mean monthly precipitation (mm) & the number of rainy days in a month respectively, and LAIm stands for the mean monthly leaf area index (m^2/m^2).

4.3.5. Calculation of ETGreen and ETBlue

The spatial and temporal availability and uses of green and blue water vary from region to region and even within a basin and have different uses (Liu and Yang, 2010). Both the blue and green waters contribute mutually to agriculture and hence assessment of their consumption (ETGreen and ETBlue) is essential for improved management of water resources in agriculture and food security in the basin. ETBlue is also known as incremental ET or secondary ET (Hoogeveen et al., 2015; Simons et al., 2020; Van Dijk et al., 2018) and ETGreen as the ET occurred due to the fraction of water infiltrates to soil and available for the plant growth (Jensen and Allen, 2016). Note that such information is not commonly available and the ability of WA+ to provide it by following a scientific procedure is a useful strength. ETGreen and ETBlue fractions are mainly used to distinguish between the consumption of precipitation versus groundwater and surface water (or stored water, viz. rivers, lakes, ponds, etc.) (Simons et al., 2020; Singh et al., 2021). The Budyko curve represents the empirical relationship between the ET, PET & P (Sposito, 2017) by representing the ratio of ET and P as Evaporative Index (EI) and PET and P as Dryness Index (DI). The Budyko curve is highly altered (i.e., the points (EI and DI)) due to extensive human interventions, particularly, due to irrigation, where the P is not the only source to meet consumptive demands (Chen et al., 2020; Simons et al., 2020).

Generally, ET/P should not exceed unity until and unless additional water is added to the basin. The Water balance approach and Energy balance approaches are incorporated into the Budyko hypothesis. A separate water balance calculation is done for each of the green and blue pixels, which identifies the rainfed and irrigated areas. Water-limited catchments ($\phi > 1$) and energy-limited catchments ($\phi < 1$) are separated by this notion (Singh et al., 2022) (Fig. 7). When the ET needs are met via rainfall, the pixels lying underneath the dotted curve are nothing but the green pixels (Singh et al., 2021) and the resulting ET is the ETGreen. If the ET is appreciably higher than the rainfall obtained when a pixel falls above the dotted curve, then it is argued that there is a further supply of water (withdrawal from groundwater or surface water) in that particular pixel and the resulting ET is the ETBlue and the incremental yield is due to the incremental (irrigation) water. The Budyko equation is expressed (Creed et al., 2014; Singh et al., 2022)as:

$$F(\phi) = \sqrt{\phi \bullet \tanh\left(\frac{1}{\phi}\right) \bullet (1 - e^{-\phi})} \tag{6}$$

$$ET_{green} = \min(F(\phi)xP, ET) \tag{7}$$

$$\phi = \frac{PET}{P} \tag{8}$$

The relationship between ETGreen and ETBlue is expressed as (Simons et al., 2020):

$$ET = ETGreen + ETBlue \tag{9}$$

In this study, the Budyko hypothesis (BH) (Budyko, 174) is used in WA+ Framework to compute ETGreen and ETBlue for irrigated and rainfed areas for the Mahi basin.

4.3.6. Estimation of land productivity and water productivity

As discussed above, we used remote sensing satellite products such as NPP, GPP, and estimates of ETGreen and ETBlue to assess the spatial and temporal variability of LP and WP. Here, we define the WP as the ratio of harvested crop yield (above-ground biomass) (dry matter) (Eq.14) per unit of the total water consumed (ET=ETGreen+ETBlue), individually for both rainfed and irrigated agriculture. Similarly, LP is defined as the ratio of harvested crop yield (dry matter) per unit of the crop acreage (Eqs. 10, 11, 12). Here we use crop-specific coefficients (harvesting index and moisture content) to convert total dry matter production to crop yield (Peiser and Bastiaanssen, 2015; Salvadore et al., 2020).

$$\text{Yield (LP)} = \frac{HI * NDM}{1 - \theta} \tag{10}$$

$$f = \frac{ET_{Blue}}{ET_{Blue} + ET_{Green}} \tag{11}$$

$$LP_{irr} = LP * f \tag{12}$$

$$LP_{rain} = LP - LP_{irr} \tag{13}$$

$$WP = \frac{LP}{ET_{Blue} + ET_{Green}} \tag{14}$$

$$WP_{rain} = \frac{LP_{rain}}{ET_{Blue} + ET_{Green}} \tag{15}$$

$$WP_{irri} = \frac{LP_{irri}}{ET_{Blue} + ET_{Green}} \tag{16}$$

where HI represents the harvest index, f is the blue water fraction and Θ denotes the moisture content (%). Yield or LP in (kg/ha) and WP in (kg/m^3). WPirri and WPrain are the WPs of the irrigated rainfed crop in (kg/m^3, respectively. Yieldirr and Yieldrain are irrigated and rainfed yields of crops in kg/ha respectively.

These estimates will help diagnose the causes of low performance and identify interventions for improving WP and LP. Worth mentioning Safi et al. (2022) that 'observing the variability of WP and LP alone will not contribute to improvements and these findings must be translated to 'actionable information by identifying the causes for the variability by the stakeholders and policymakers. A representative crop or crops are to be chosen to compute the fresh crop yield for each of the agricultural crop classes. Here we choose the representative crop (cereals, which mainly include rice and wheat) as the major crop class in the basin. For each type of crop, harvest indices and water content are calculated to translate dry matter output into crop yield aiding growing season influence (Singh et al., 2021). According to the Indian crop calendar, Rabi and Kharif seasons are considered as the growing seasons of irrigated and rainfed cereals.

4.3.7. Performance indicators

WA+ includes several standard indicators. Each sheet has its own set of indicators that are derived from the data in the sheets as given in Table 6 (Section 4.2.7) to comprehend the water consumption patterns in the basin and devise various management strategies.

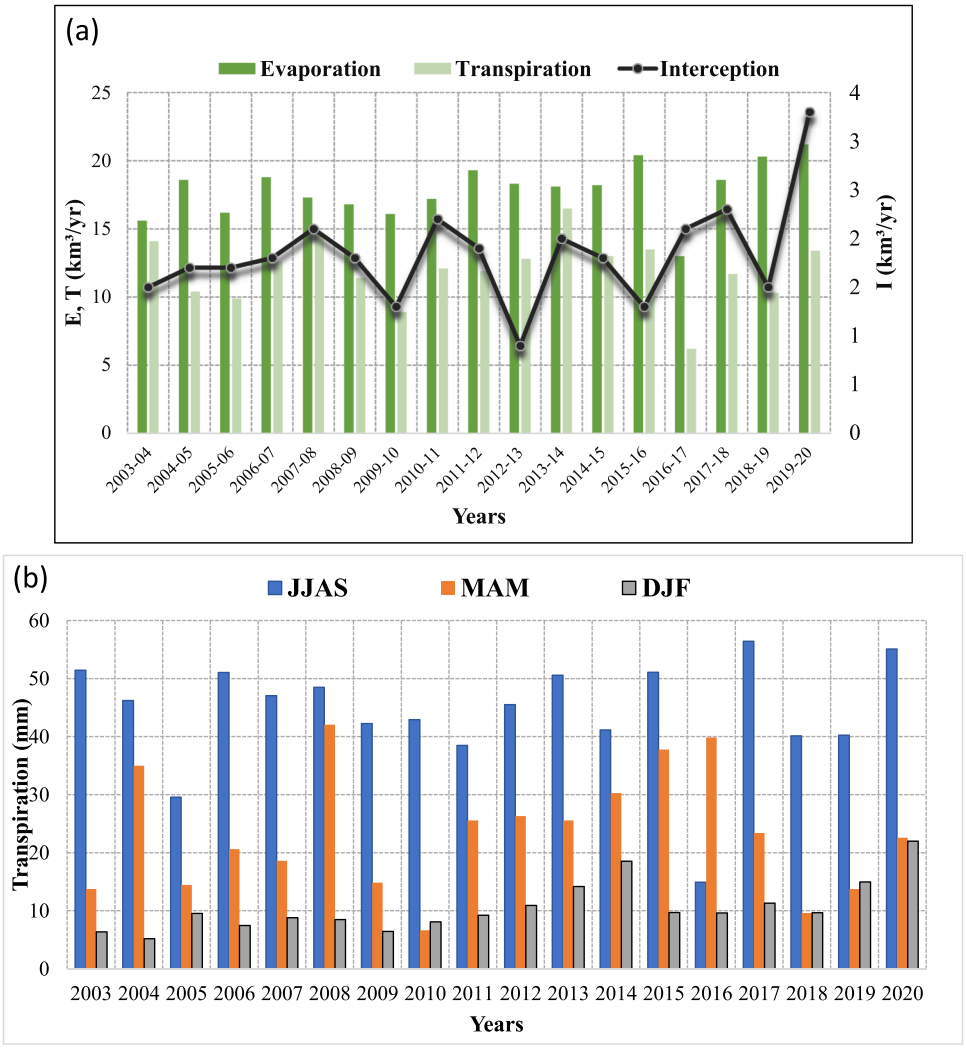


Fig. 8. a: Annual variation of E, T, and I for the period of 2003–2020, b: Seasonal variation of E, T, and I for the period of 2003–2020, where JJAS represents june-july-august-september, MAM shows march-april-may and DJF shows december-january-february.

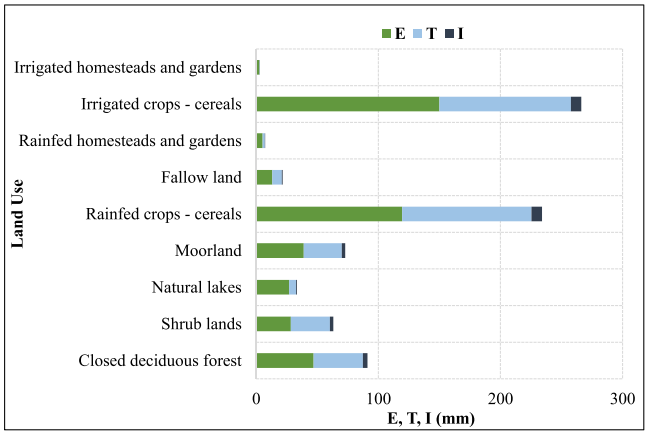


Fig. 9. Influence of the land use classes on the E, T, and I.

5. Results and discussion

This section presents the results obtained from this study and critically discusses these results and their implication in the field applications for improving agricultural water management and enhancing WP and LP for managing water resources and food security in the basin.

5.1. ET partitioning and land use interaction

The knowledge of the water fluxes (ET and its components E, T, and I) within the soil-vegetation system to maximize the productive water loss (T) and minimize the non-productive water losses (E and I) is crucial (Sutanto et al., 2012). The ET partitioning is also a crucial factor for global climate models as the partitioning of ET has proven to be a significant source of uncertainty for future climate projections (Lawrence et al., 2007). However, the quantification of the relative contribution of E and T from ET (Zhang et al., 2010) is difficult. In this work, the WA+ framework is used for the partitioning of ET considering the interaction of distinct land and water use categories, i.e., PLU, ULU, MLU, and MWU. The ET partitioning as per four major landuse classes will further enrich its usefulness in agriculture. ET partitioning as per the four land and water use classes was done using the WA+ framework (as discussed in Section 3.7) and the results are given in Table 2. The results show that the long-term average annual ET in the Mahi basin is 27.48 km^3 /year with partitioning into E, T, and I for all four land and water use classes for the period of 2003–2020. This specific partitioning with respect to land and water use classes is extremely useful for managing agricultural waters and ecosystems. The ULU (consists of closed deciduous and green forest, moorland, and shrubs, which is the case

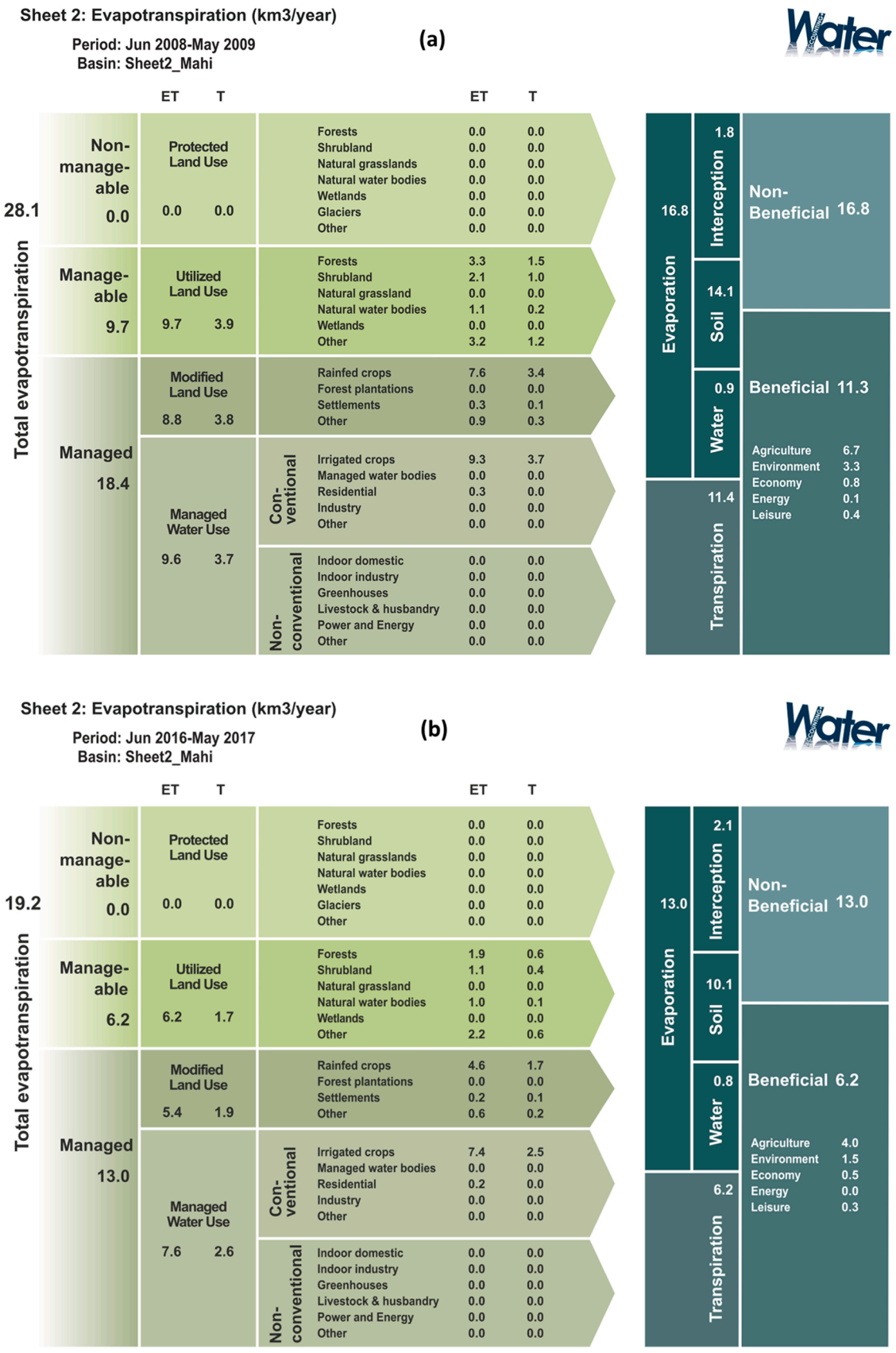


Fig. 10. The 'evapotranspiration' sheets for (a) dry-Year (2008–09) and, (b) wet-Year (2016–17).

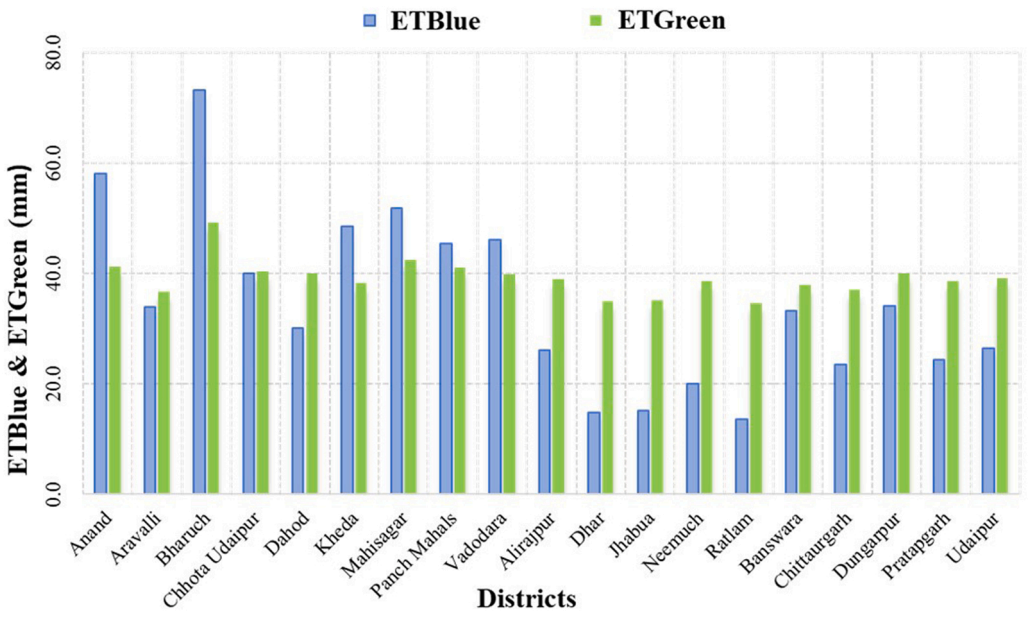


Fig. 11. District-wise long-term average of ETBlue and ETGreen.

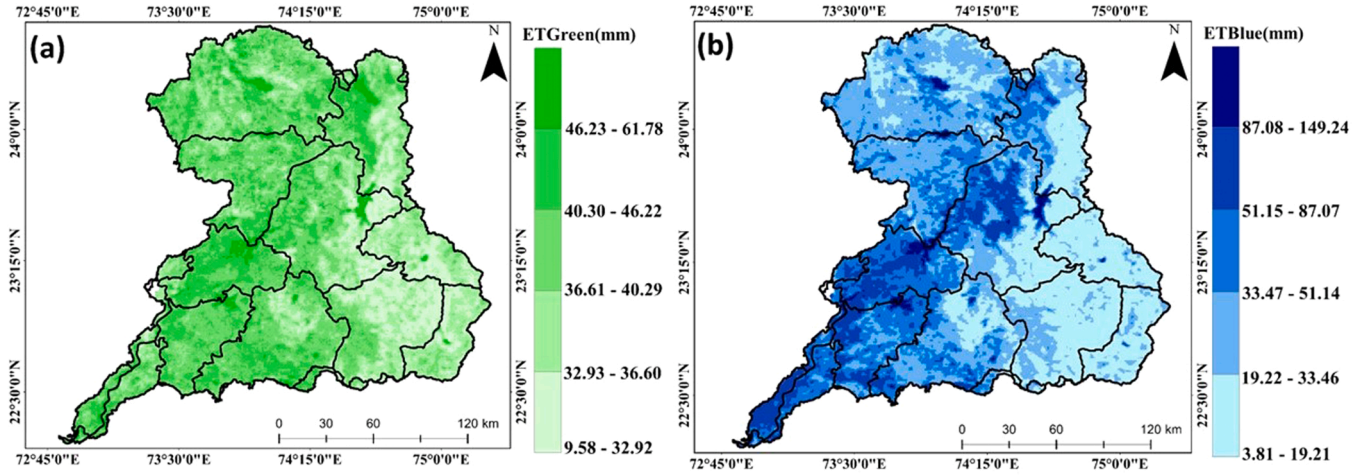


Fig. 12. Spatial variation of annual average (2003–2020) (a) ETGreen, and (b) ETBlue

with the Mahi basin) contributes maximum to ET, i.e., 35.30 % (9.36 km^3/year). Surprisingly, similar is the case with the MWU class (irrigated crops mainly), i.e., it contributes 32.63 % (9.23 km^3/year). The MLU class (rainfed agriculture mainly) contributes 31.32 % (9.01 km^3/year) of the total ET. Agriculture contributes about 64.62 % of the total ET in the Mahi basin. The ET partitioning shows that the T for MWU is lower as compared to the MLU class, which indicates that there exists a scope for agricultural water management (maybe by using improved crop varieties or by improved irrigation practices or both) in this basin to increase beneficial water consumptions. The E component (from soil and water both) in MWU classis higher as compared to the MLU class. Table 2 also shows that though the area under MWU is less as compared to the MLU class, but the E component (non-beneficial) is higher. Fig. 8a also shows the variability of ET components (E, T, and I) for the period of 2003–2020. This shows that the E component is always higher than the T and has an increasing pattern for most of the years, except for a few years. Fig. 8b represents the seasonal variation of transpiration. Fig. 9 shows the variation of ET components as per the landuse classes. A close look at Fig. 9 shows that the irrigated crops have a higher E component than the rainfed crops followed by the ULU class.

The ET partitioning also shows that the average annual T of the basin is 11.33 km^3 /year, which is about 40.16 % of the average annual ET, whereas the E component (from soil and water) is about 15.93 km^3 / year (56.44 %) and accounts for 0.96 km^3 /year (3.40 %) of the ET. The T component is referred to as the beneficial ET because it is responsible for food and fiber production and contributes to the economy. The T component is highest in the MLU followed by the MWU and ULU, which implies that rainfed agriculture contributes significantly to the food security of the basin. Similar results were also reported by (Singh et al., 2022) while estimating evaporative demand using the Budyko hypothesis for the Mahi basin. Overall, these findings imply that there is an enormous possibility to put into effective water management practices in the basin which will reduce non-beneficial consumption (E and I) along with an increase in beneficial consumption via advanced irrigation practices. The water thus saved may be used for irrigating larger areas or for other purposes than agriculture. Fig. 10(a-b) also shows the ET-partitioning sheet developed by WA+ Framework for the Mahi basin, respectively for wet and dry years. These sheets can assist in formulating 'actionable' plans (agronomical, engineering, and societal-managerial) for reducing E and I and increasing T from irrigated crops (in particular) and rainfed in general.

Table 3
District-wise long-term average of ETBlue and ETGreen and BWI and GWI.

Districts	ETBlue (mm/month)	ETGreen (mm/month)	(BWI)	GWI
Anand	58.1	41.3	0.58	0.42
Aravalli	34.0	36.7	0.48	0.52
Bharuch	73.2	49.2	0.59	0.40
Chhota Udaipur	40.1	40.4	0.49	0.50
Dahod	30.2	40.0	0.43	0.57
Kheda	48.5	38.4	0.55	0.44
Mahisagar	51.9	42.5	0.55	0.45
Panch Mahals	45.4	41.0	0.52	0.47
Vadodara	46.1	39.9	0.53	0.46
Alirajpur	26.2	39.0	0.40	0.59
Dhar	14.9	34.9	0.29	0.70
Jhabua	15.2	35.1	0.30	0.69
Neemuch	20.1	38.6	0.34	0.66
Ratlam	13.6	34.6	0.28	0.72
Banswara	33.2	38.0	0.46	0.53
Chittaurgarh	23.5	37.1	0.38	0.61
Dungarpur	34.1	40.1	0.46	0.54
Pratapgarh	24.4	38.6	0.38	0.61
Udaipur	26.5	39.2	0.40	0.59

Table 4
Estimates of the LP and WP of the rainfed & irrigated rice for period of 2003–2020.

Year	Yield rainfed [kg/ha]	Yield irrigated [kg/ha]	WP Irrigated [kg/m^3]	WP rainfed [kg/m^3]
2003–04	1850.31	2235.18	0.87	0.42
2004–05	1541.22	2148.42	0.81	0.38
2005–06	1762.85	2247.10	0.80	0.52
2006–07	1598.62	2290.82	0.81	0.35
2007–08	1717.58	2092.92	0.65	0.39
2008–09	1777.69	2033.88	0.78	0.44
2009–10	1717.50	1900.80	0.78	0.46
2010–11	1651.24	2520.42	0.94	0.38
2011–12	1602.36	2362.73	0.70	0.36
2012–13	1541.47	2450.93	0.71	0.35
2013–14	1868.01	3192.73	0.73	0.39
2014–15	1749.29	2293.30	0.54	0.43
2015–16	1703.83	1738.49	0.42	0.48
2016–17	1788.98	2524.93	0.91	1.54
2017–18	1841.56	1962.00	0.61	0.47
2018–19	1587.57	2216.12	0.65	0.41
2019–20	1832.32	2680.21	0.54	0.44
Average	**1713.67**	**2287.71**	**0.721**	**0.483**

5.2. Estimation of ETGreen and ETBlue

We know that green water is the sole water source for rainfed agriculture and an important part of the water source for irrigated croplands (Li et al., 2022). Agriculture is the only sector where green and blue water both contribute mutually (Liu et al., 2022) and that crop irrigation consumes the largest proportion of blue water (Mekonnen and Hoekstra, 2012). (Mao et al., 2020) state that even though green water is the dominant water resource, blue water is also critical which is back into the soil through physical and human-induced processes to meet the water demand of the ecosystem. In this study, ETBlue and ETGreen are quantified using the Budyko hypothesis. The results show that the long-term average annual ETGreen in the basin is 15.8 km^3 /year which is 1.28 times higher than the ETBlue (12.32 km^3/year). The district-wise estimates of the long-term monthly average of ETGreen and ETBlue are shown in Fig. 11 and the spatial variability is shown in Fig. 12 (a-b). In general, the results show that the districts falling in the Gujarat state have more ETBlue followed by the districts in Rajasthan and Madhya Pradesh. Within the Gujarat state, the Bharuch (73.2 mm/month) and Anand (58.1 mm/month) districts have the largest ETBlue followed by the Mahisagar district (51.9 mm/month) than the others. Fig. 12(a-b) also shows that the districts falling in the states of Madhya Pradesh and Rajasthan in the Mahi basin have higher ETGreen than ETBlue). The district-wise estimates of ETGreen and ETBlue are also given in Table 3. Both ETGreen and ETBlue are lowest in Ratlam (34.56 mm/month & 13.63 mm/month) district and highest in the Bharuch (49.28 mm/month & 73.20 mm/month) (Table 3) district of the Mahi basin. We also calculated the blue water index (BWI = ETBlue/(ETBlue+ETGreen)) and green water index (GWI = ETGreen/(ETBlue+ETGreen)) , and the results are given in Table 3. An index value of 1 shows that the crop yield is from irrigation only, whereas an index of 0 indicates that precipitation is sufficient to achieve maximum crop yields (Johansson et al., 2016).

The BWI and GWI are found to vary from 0.282 to 0.598 and 0.40–0.72. The BWI is highest in the district of Gujarat and the GWI is highest in the districts of Madhya Pradesh. The higher value of BWI is an indication of the intensively irrigated agriculture in the basin districts. The districts like Bharuch, Anand, Kheda, Mahisagar, Panchmahal, and Vadodara are more dependent on the irrigation for their agricultural production than the other districts. This the quantification of ETBlue and ETGreen (both spatial and temporal) and will help identify the BWI hotspots (districts) for managing water in agriculture by adopting advanced irrigation technologies or by using less water-requiring crops in those districts.

5.3. Spatial and temporal variability of LP and WP

It is crucial to quantify the crop yields, water consumption, and WP for an efficient agricultural water management and sustaining food security in a basin. The spatial and temporal quantification of these indicators helps identify the 'bright spots' and 'hot spots' to devise 'actionable goals. The bright spots (well-performing crop fields) are the 'regions' or 'spots' with the highest WP and LP and the hot spots (poor-performing crop fields) are the 'regions' or 'spots' with the lowest WP and LP. Due to the complex land use land cover and hydrological processes involved, the quantification and assessment of WP and LP are usually difficult, especially at the regional scale (Ren et al., 2021). These assessments require an understanding of the relationship between the hydrological variables such as E, T, I, and other bio-physical variables (Karimi et al., 2013; Safi et al., 2022) such as LAI, NPP, and GPP. Secondly, at the same time, the field experiments and the regional observations are also scanty and are usually time-consuming to collect the required data.

As discussed in the previous sections, these limitations can be sorted out by using the newly developed WA+ framework which translates open-source earth observation datasets to LP and WP for rainfed and irrigated agriculture individually using the Budyko hypothesis and establishes the linkage between the hydrological and biophysical variables. Following this, the WA+ framework is used for quantifying the WP and LP for the Mahi basin at the district administrative scale. The mathematical expressions for the estimation of LP and WP from irrigated and rainfed agriculture are discussed in Section 4.2.6. WA+ reports these results in Agricultural Services or Productivity sheet. The spatial maps of the WP and LP are also developed in the WA+ framework which is very useful for identifying the bright spots and hot spots in the basin. In the Mahi basin, the cereals are the major crop of the basin and are grown in both the seasons of Rabi and Kharif. As mentioned in Section 4.2.6, we took the representative crop as cereals to compute the LP and WP and other indicators. The results show that the long-term average LP of the irrigated and rainfed cereals is 2287.71 kg/ha & 1713.62 kg/ha, respectively from 2003 to 2020 (Table 4). Similarly, the long-term average of WP of the irrigated and rainfed cereals is 0.721 kg/m^3 & 0.483 kg/m^3 , respectively from 2003 to 2020. Fig. 13(a-b) (agricultural services sheets) show the estimates of LP and WP for the year 2008–09 and 2016–17, individually (as well as average) for both therainfed and irrigated agriculture. A close look at the agricultural services sheets (Fig. 13a-b) shows that in case of the irrigated agriculture, the

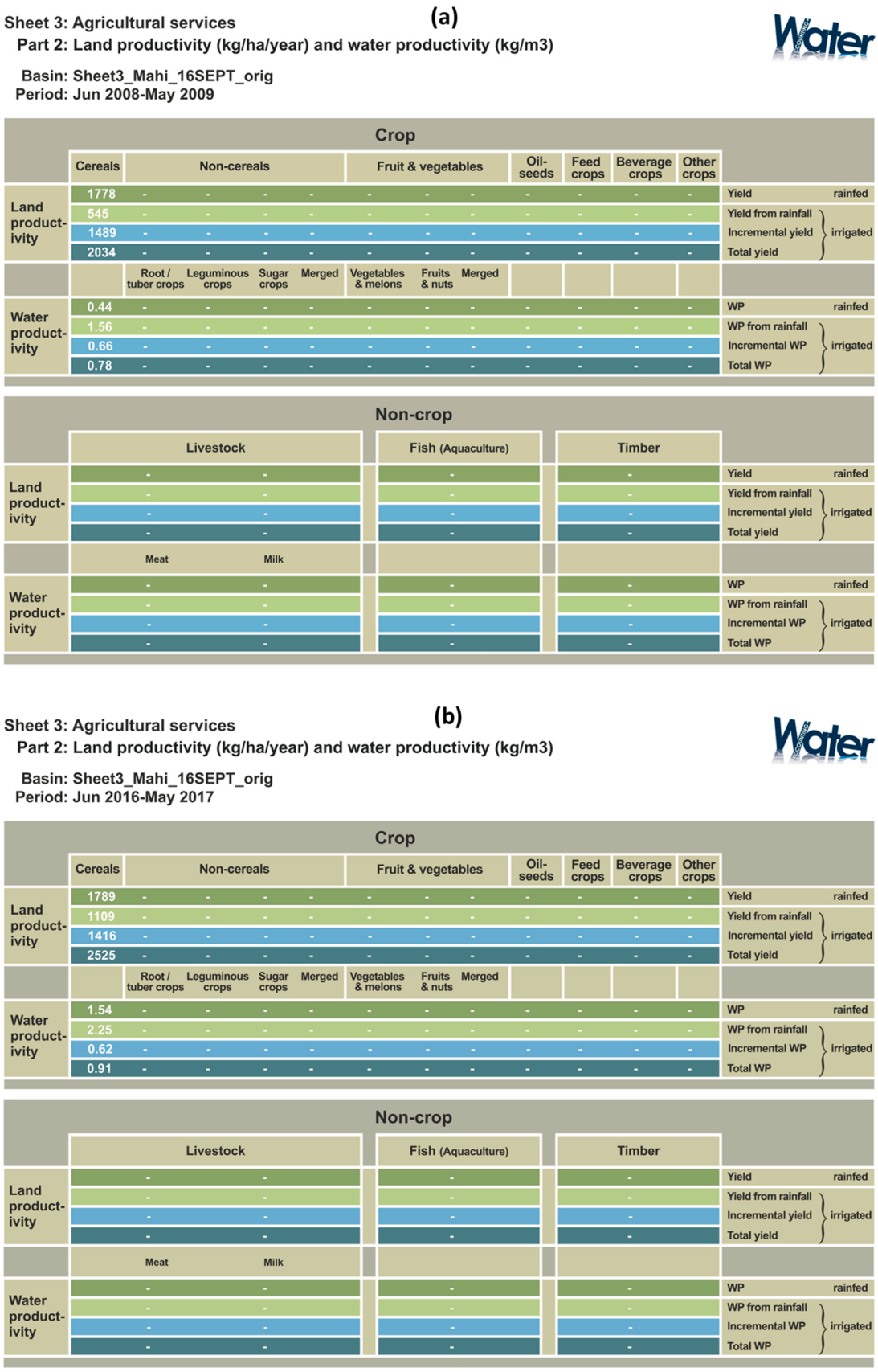


Fig. 13. Agricultural Services sheet for (a) Dry Year (2008–09), (b) Wet Year (2016–17), showing the Land Productivity (Yield or LP) and Water Productivity (WP) of rainfed and irrigated cereals.

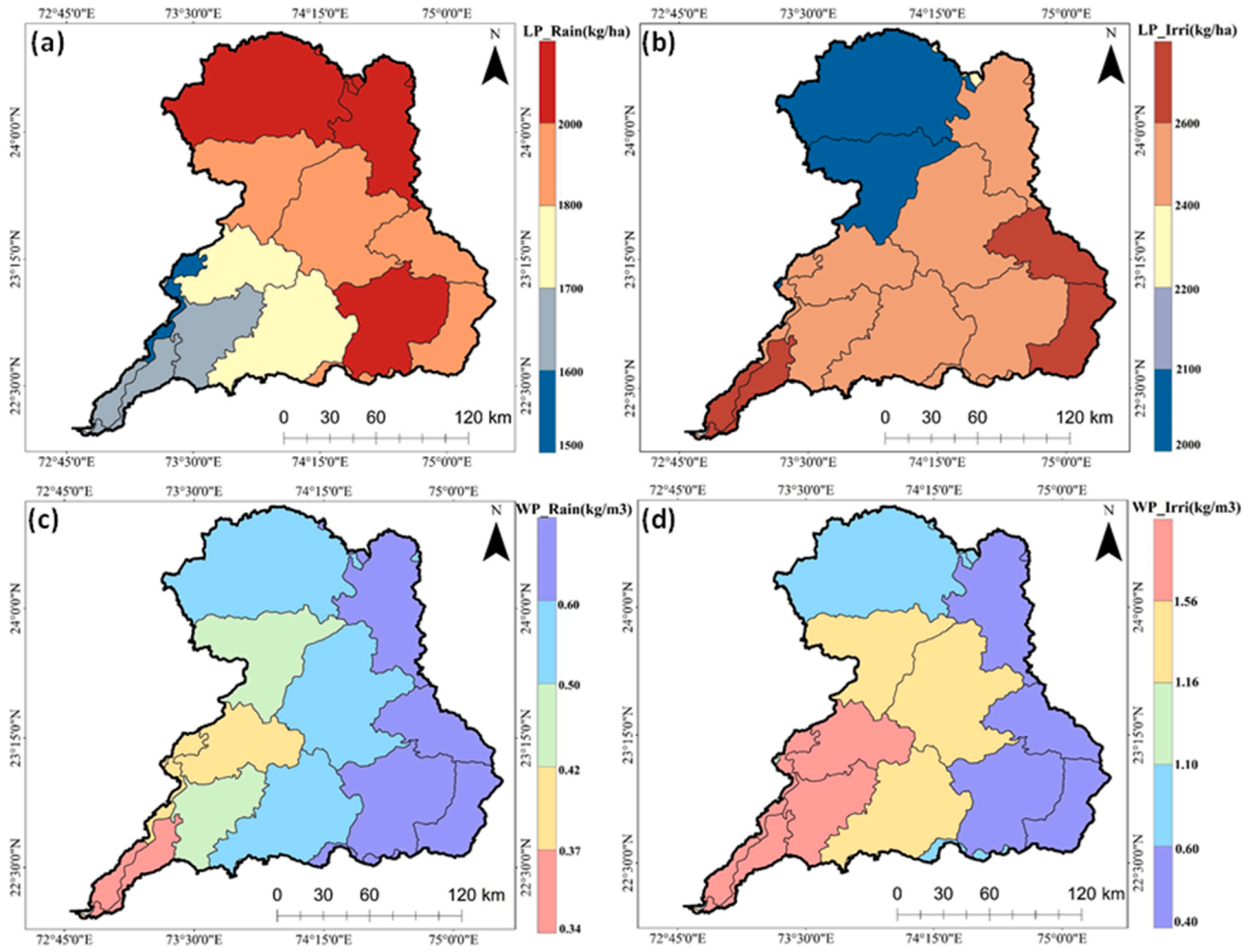


Fig. 14. Spatial variation in annual average (2003–2020) (a) LP of Irrigated crops, (b) LP of Rainfed crops, (c) WP of irrigated crops and, (d) WPof rainfed crops.

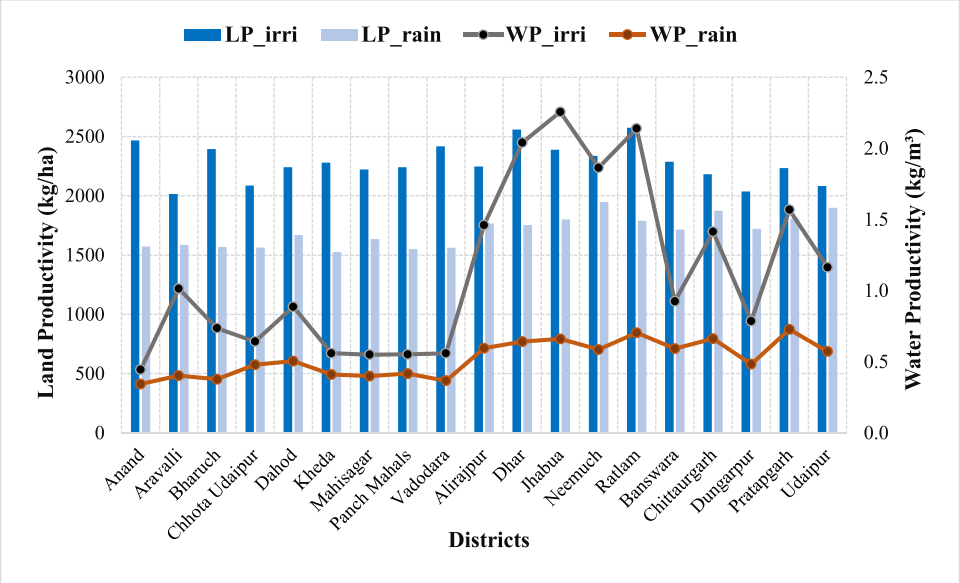


Fig. 15. District-wise average LP and WP of irrigated and rainfed crops.

estimation of the LP and WP is due to both rainfall and incremental irrigation, whereas, in case of the rainfed agriculture, the estimates of LP and WP are due to rainfall only.

Fig. 14 a-d shows the spatial variability (district-wise) of WP and LP for rainfed and irrigated cereals in the Mahi basin. Fig. 15 shows the comparison of the WP and LP (irrigated and rainfed both) for all the

Table 5
District-wise LP (Yield) and WP of rainfed and irrigated crops.

District	LP_rain (kg/ha)	LP_irri (kg/ha)	WP_rain (kg/m^3)	WP_irri (kg/m^3)
Anand	1572.21	2467.51	0.34	0.44
Aravalli	1585.81	2014.61	0.40	1.01
Bharuch	1568.56	2394.21	0.38	0.74
Chhota Udaipur	1564.15	2086.94	0.48	0.64
Dahod	1668.81	2241.13	0.50	0.89
Kheda	1526.79	2279.85	0.41	0.56
Mahisagar	1635.45	2222.60	0.40	0.55
Panch Mahals	1551.66	2241.43	0.42	0.55
Vadodara	1561.28	2416.77	0.37	0.56
Alirajpur	1766.53	2247.03	0.60	1.46
Dhar	1753.62	2558.22	0.64	2.04
Jhabua	1801.69	2387.65	0.66	2.26
Neemuch	1948.13	2337.66	0.59	1.86
Ratlam	1790.09	2573.96	0.70	2.14
Banswara	1715.34	2287.68	0.59	0.93
Chittaurgarh	1873.48	2182.37	0.66	1.41
Dungarpur	1720.05	2037.34	0.48	0.79
Pratapgarh	1808.16	2233.03	0.73	1.57
Udaipur	1898.93	2082.35	0.57	1.16

Table 6
Performance Indicators.

Indicator	Formulation	Value	Remarks
Transpiration ET fraction	$\frac{\text{Transpiration}}{\text{ET}}$	0.40	Low vegetation water consumption
Managed ET fraction	$\frac{\text{ETmanaged}}{\text{ET}}$	0.66	Human activities controlling ET
Agricultural ET fraction	$\frac{\text{Agricultural ET}}{\text{ET}}$	0.61	Human activities controlling ET
Irrigated ET fraction	$\frac{\text{Irrigated ET}}{\text{Agricultural ET}}$	0.55	Agriculture is major water consumer
Food-irrigation dependence	$\frac{\text{Irrigated agriculture ET}}{\text{Total Agriculture ET}}$	0.90	Food-security is relying on the Irrigation

districts of the Mahi basin. The district-wise long-term average of the LP and WP is also given in Table 5. It can be inferred from Fig. 15 and Table 5 that the districts such as Anand, Bharuch, Vadodara, Dhar, Neemach, Jhabua, and Ratlam have higher LP (irrigated) (more than 2300 kg/ha) with the highest value for Ratlam (2573.96 kg/ha) than the other districts of the basin. The LP (irrigated) is lowest for the Aravali district of Rajasthan than the other districts. The districts falling in the state of Madhya Pradesh i.e., Neemach, Jhabua, Alirajpur, and the districts falling in the state of Rajasthan, i.e., Udaipur Chhataurgarh, Pratapgarh, and others have higher LP (rainfed) than all the districts of Gujarat. The Neemach district of Madhya Pradesh is found to have the highest LP (rainfed) (1948.13 kg/ha) followed by Udaipur (1898.93 kg/ha) with the lowest value of LP for the Kheda district (1526.79 kg/ha) of Gujarat. Based on the LP estimates, it is concluded that the Ratlam and Neemach districts of Madhya Pradesh have the highest LP for both irrigated and rainfed cereals. Fig. 14c-d, Fig. 15 and also show the estimates of WP of the cereals (irrigated and rainfed both) for all the districts of the Mahi basin. The WP (rainfed) of all the districts of Gujarat is comparatively lower (varying from 0.34 kg/m^3 to 0.5 kg/m^3) than the districts of Madhya Pradesh (varying from 0.59 kg/m^3 to 0.70 kg/m^3) and the Rajasthan (varying from 0.48 kg/m^3 to 0.73 kg/m^3). The Pratapgarh district of Rajasthan has the highest rainfed WP (0.73 kg/m^3) followed by Ratlam (0.70 kg/m^3) of Madhya Pradesh. The WP (irrigated) for all the districts of the Mahi basin is also given in Fig. 14c-d, Fig. 15 and Table 5. The WP (irrigated) of all five districts of Madhya Pradesh is found to vary from 1.46 kg/m^3 to 2.26 kg/m^3 followed by the districts of Rajasthan (varying from 0.79 kg/m^3 to 1.57 kg/m^3) and Gujarat (0.44 kg/m^3 to 0.89 kg/m^3). These estimates of WP and LP (irrigated and rainfed both) can be further used for the classification of the districts of the Mahi basin in 'bright spots-district' and 'hot spot-district using the criteria given by Safi et al. (2022). For irrigated cereals, the districts falling in the state of Madhya Pradesh have the highest LP and WP than the states of Rajasthan and Gujarat. The Ratlam and Dhar districts report the highest WP (irrigated) and LP (irrigated) among all the districts and Ratlam with the highest values of both WP and LP, i.e., 2.14 kg/m^3 and 2573.96 kg/ha, respectively. Hence Ratlam district is classified as the 'Bright spot-district'. Similarly, the Anand district is found to have the lowest WP (irrigated) and is classified as the 'hot spot-district'. As for as rainfed agriculture is concerned, the districts falling in the state of Madhya Pradesh have the highest WP and LP followed by Rajasthan and Gujarat. The Neemuch district of Madhya Pradesh is classified as the 'Bright spot-district' with WP and LP as 0.59 kg/m^3 and 1948.13 kg /ha, respectively, and Anand as the hot spot- district with the lowest WP and LP as 0.34 kg/m^3 & 1572.21 kg/ha, respectively. Overall, at the state level (region level), it can be stated the state of Madhya Pradesh is the 'Bright Spot-state' and Gujarat as the 'Hot spot-state' in irrigated and rainfed agriculture both.

5.4. Performance indicators

The basin performance indicators such as transpiration fraction (T/ET) (Cao et al., 2022; Tong et al., 2019), beneficial ET fraction, Agricultural ET fraction, Irrigated ET fraction, and Food-Irrigation dependency (Karimi et al., 2013a,b)() plays an important role in assessing how do the water is being used in producing services such as in agriculture and sustaining ecosystem services. Along with this, these indicators also indicate the coupling characteristics of the ecological and hydrological systems in a basin (Tong et al., 2019). Worth mentioning Cao et al. (2022) that the transpiration ratio (T/ET) also plays an important role in understanding the water cycle and land-atmosphere interactions (Kool et al., 2014). The ratio T/ET has great scientific and practical importance in sustainable development in the future (Tong et al., 2019). These works show the importance of basin indicators for sustaining water resources and food security in a basin. The estimated values of these indicators are given in Table 6. The ratio (T/ET) also gives the fraction of ET vaporized via plant leaves. The managed ET fraction represents gives the fraction of ET through human interventions, i.e., through MLU and MWU classes and help may help in saving water in agriculture. The agricultural ET reflects how intensively the water is being used in agriculture and irrigated ET fraction reflects the contribution of irrigated agriculture to the ET. Finally, the food-irrigation dependency reflects the dependency of the basin on irrigated agriculture for food production.

The long-term average (for the period of 2003–2020) of all indicators is shown in Table 6. The higher the T/ET fraction better is the performance of the basin. and for the Mahi basin, it is 40 %, and this indicates that there is a significantly lower amount of vegetation water consumption, which in turn indicates that most of the water in the basin is depleted due to soil moisture evaporation and this is remarkable but unfavorable. This necessitates for application of conservation practices and advanced irrigation technologies for reducing 'E′ from the basin. The managed ET fraction is 0.66 implying that human activities are controlling the water depletions largely in the basin. The agricultural ET fraction in the basin is 61 % showing that agricultural activities consume the largest amount of water. The reason for this is the basin's extremely large dependence on agricultural activities as the LULC map shows (Section 3.6) that there is 63% of agricultural land. The irrigated ET fraction for the basin is 55 %, representing that blue water (incremental ET) consumption is contributing 55% to agriculture. Food irrigation dependency is about 90 %, which shows that food security is significantly influenced by the irrigation activities in the basin. Fig. 16a shows the yearly variation of the performance indicators for the Mahi basin. Out of these five indicators, as mentioned herein and taken from (Karimi et al., 2013), only the transpiration ratio (T/ET) has been a topic of much attention among researchers. Cao et al. (2022) reported the values of T/ET varying between 0.52 and 0.61 across the world and found that

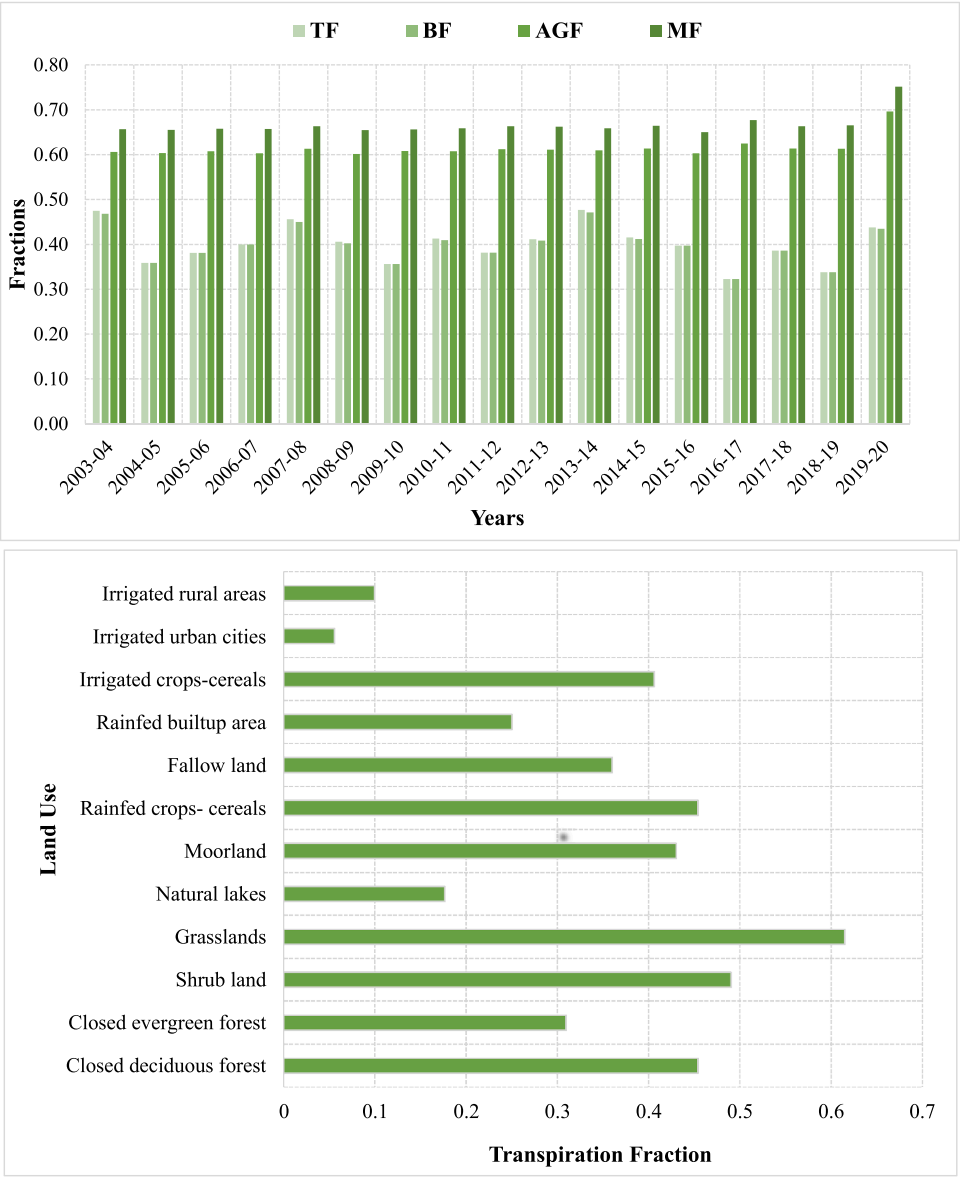


Fig. 16. (a) Annual variations of transpiration fractions (TF), beneficial ET fractions (BF), agricultural ET fraction (AGF), managed ET fractions (MF) for the period of 2003–2020, and (b) Transpiration fractions (T/ET) variation in different land uses classes.

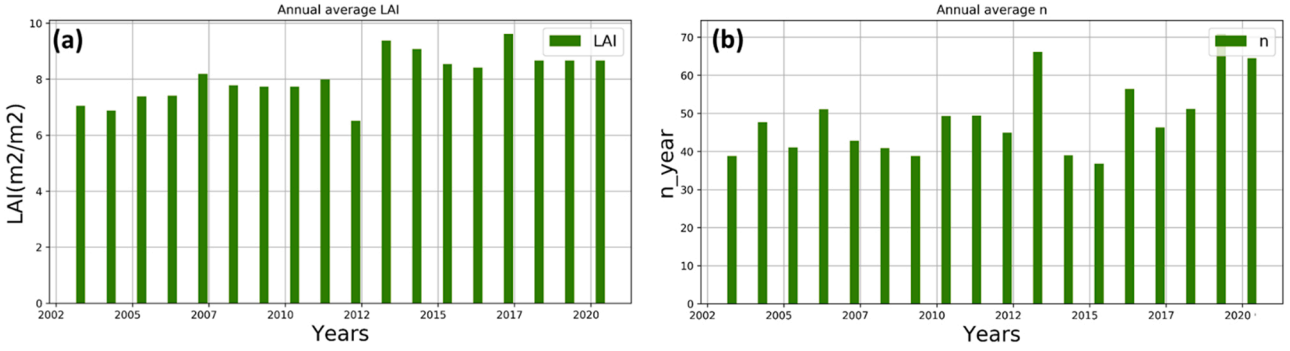


Fig. A1. (a) Annual variation of LAI from (2003–2020), (b) annual variation of number of rainy days (n) from (2003–2020).

LAI (vegetation water consumption) is the most important factor controlling it in a basin. Coenders-Gerrits et al., (2014) reported T/ET range of 0.35–0.80. Tong et al. (2019) quantified the T/ET ratio for different ecosystems and reported that the average T/ET is highest for the cropland ecosystem (0.80), followed by the alpine meadow ecosystem (0.79), the desert riparian forest (0.67), the desert riparian shrub ecosystem (0.67 ± 0.06), and meadow (0.55 ± 0.23). The LAI was found to be the first order-controlling factor on the T/ET ratio. The T/ET

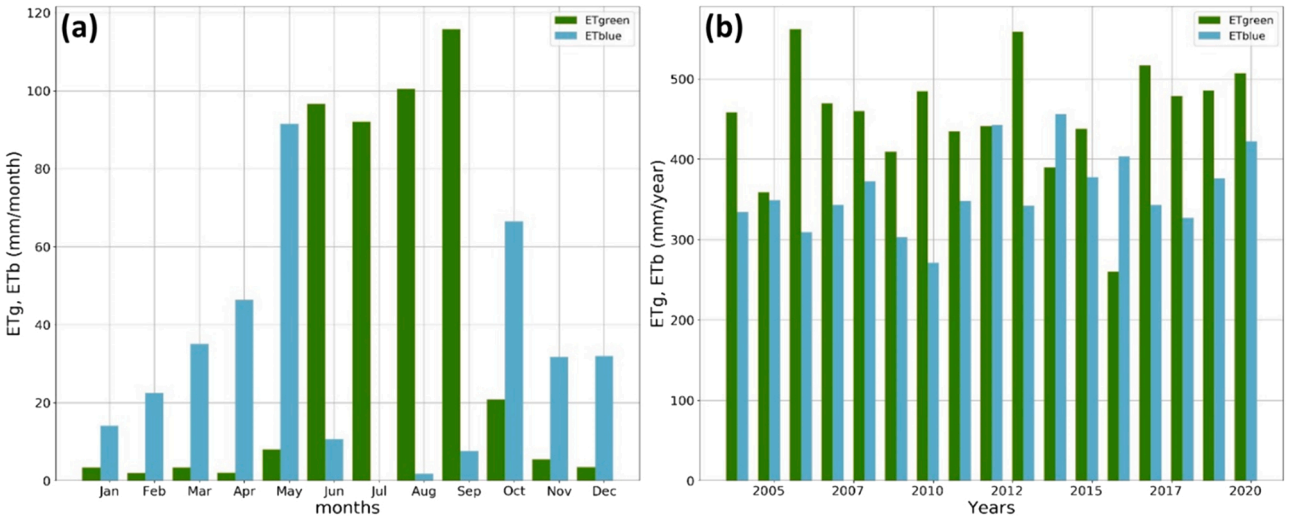


Fig. A2. : (a) Monthly variation of ETGreen and ETBlue, (b) annual variation of ETGreen and ETBlue.

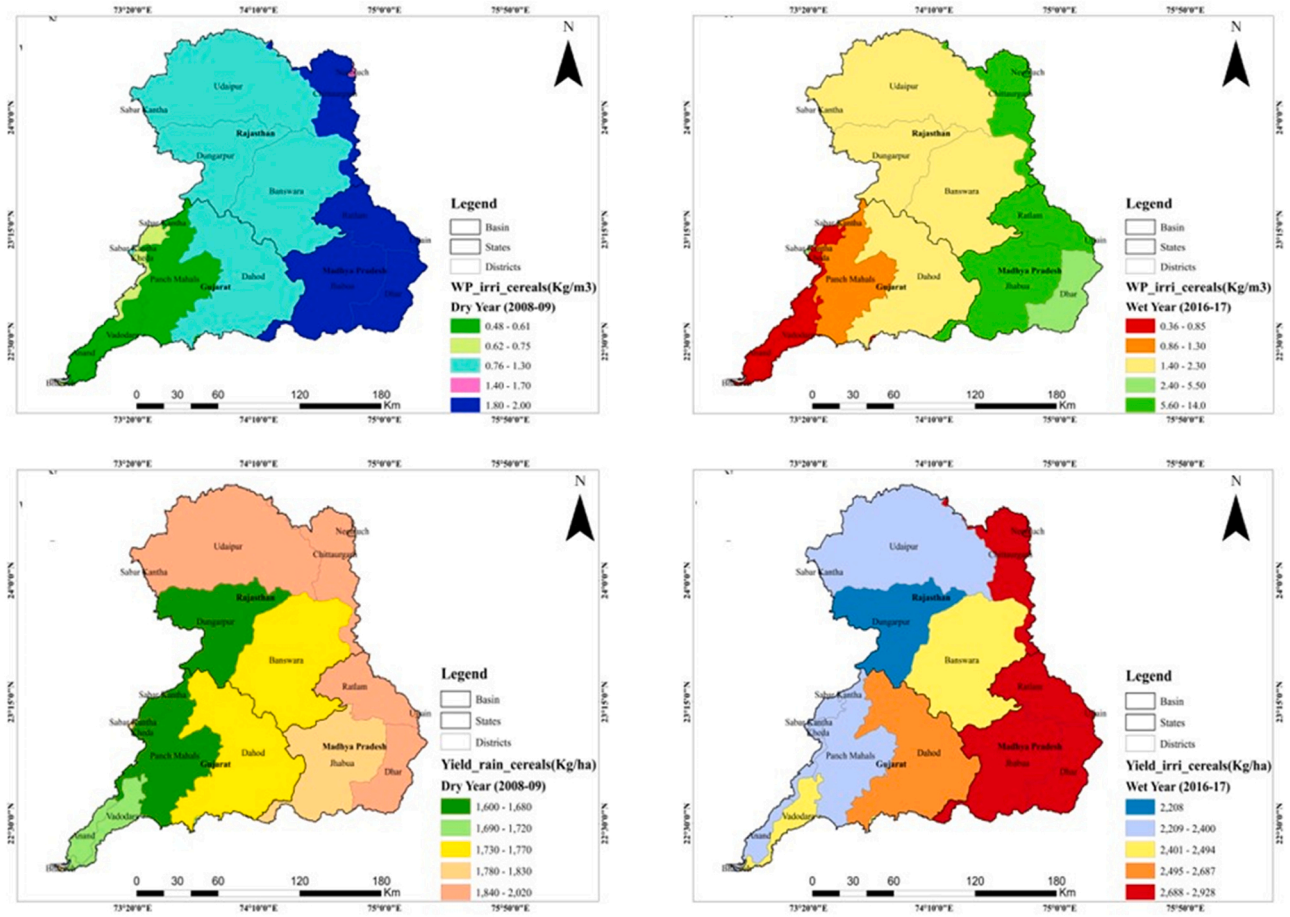


Fig. A3. : LP & WP of irrigated crop on dry & wet years.

ratio was also found to be different for different ecosystems. In this study, irrigated agriculture is found to have a lower value of T/ET ratio (0.40) as compared to the rainfed crops (0.45) as shown in Fig. 16b. This reflects there is a need to either enhance the ecosystem productivity or apply controls on ET process through reduced water supplies in terms of incremental irrigation.

The average Transpiration fraction (T/ET) of different Land uses is shown in Fig. 16b. The T/ET is found to be highest for the Grassland ecosystem (0.62), followed by shrublands (0.50), forest cover (0.45), and rainfed crops (0.45), irrigated crops (0.40) and least for the waterbodies (0.18) and built-up (0.099 $\pm$ 0.05) areas respectively. The T/ET ratio is higher for rainfed agriculture (0.45) than the irrigated agriculture (0.40), however, still it has a larger scope for improvements in both the cases, i.e., for both rainfed and irrigated agriculture. These indicators highlight the basin's performance in terms of ecosystem productivity and water services in the basin and possible improvements

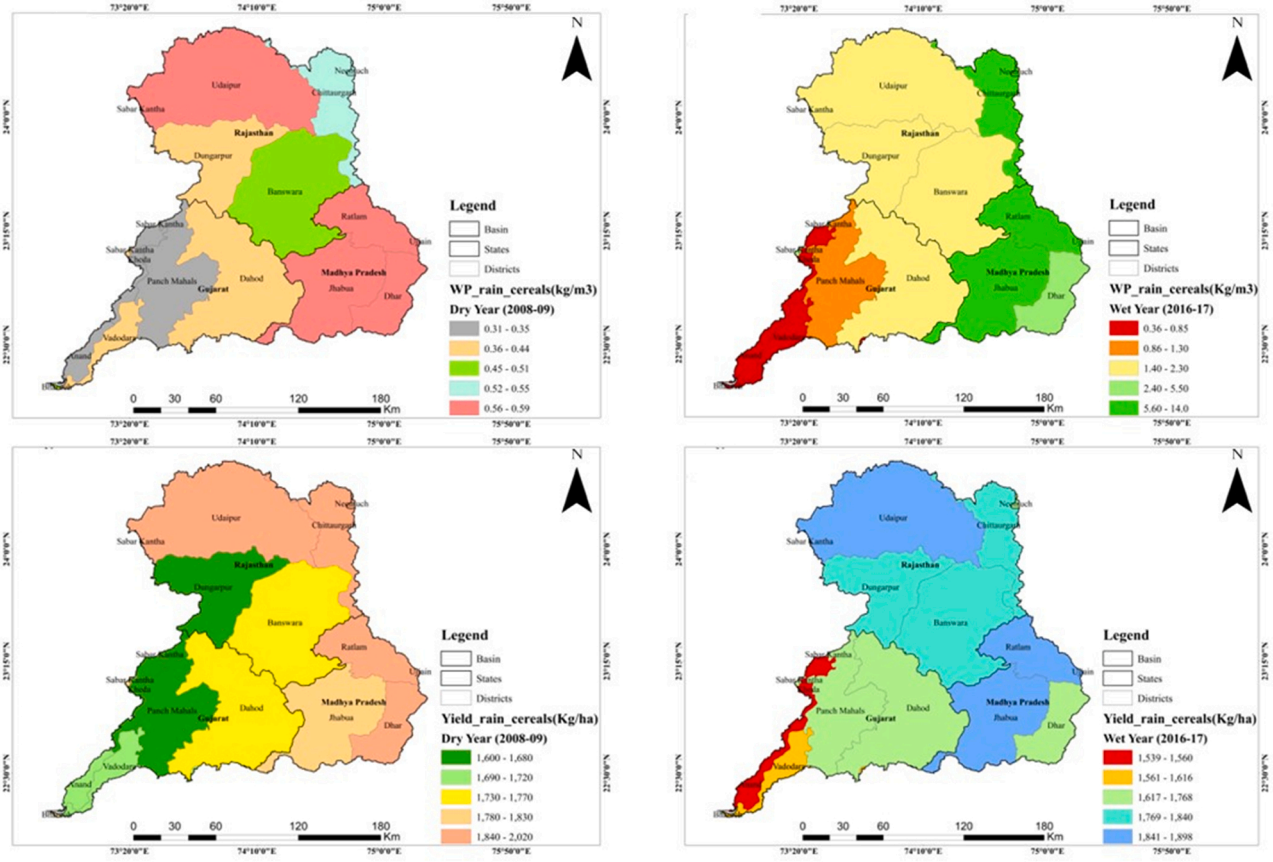


Fig. A4. : LP & WP of rainfed crop on dry & wet years.

for efficient agricultural water management and sustaining food security in the basin.

6. Conclusions

In this study, we applied the newly developed WA+ framework for estimating the water consumption patterns in terms of ETGreen and ETBlue for assessing the LP and WP in a semi-arid Mahi basin in India to formulate the "actionable management plans" for the stakeholders and policymakers. We also partitioned ET flux into E, T and I flux as per the four major landuse and water classes and quantified the basin performance indicators. For this purpose, in this study, we used freely available open-source remote sensing datasets. By assessing the spatial and temporal variability of both LP and LP at the administrative units (districts in this case) scale we classified these administrative units as 'bright spots' and 'hot spots' to help identify the causes behind what is affecting their performance in the basin. The following conclusions are drawn from this study as:

- The long-term ETGreen and ETBlue in the Mahi basin are found to be 15.8 km^3/year and 12.32 km^3/year. Respectively. In general, the districts falling in the state of Gujarat have the highest values of long-term average annual ETBlue than the administrative units falling in the state of Rajasthan and Madhya Pradesh.
- The districts of Gujarat such as Bharuch, Anand, and Mahisagar have the highest values of long-term average annual ETBlue, i.e., 73.2 mm/month, 58.1 mm/month, and 51.9 mm/month, respectively than the districts of the Rajasthan and Madhya Pradesh in the Mahi basin.
- The BWI and GWI in the Mahi basin vary from 0.28 to 0.59and 0.40–0.71. The BWI is highest in the district of Gujarat and the GWI is highest in the districts of Madhya Pradesh. The higher value of BWI is an indication of the more dependency on irrigation for agricultural production and vice versa.
- The long-term average of LP and WP of the irrigated and rainfed cereals is found to be 2287.71 kg/ha & 1713.62 kg/ha and 0.721 kg/m^3 & 0.483 kg/m^3 , respectively for the period of 2003–2020.
- The districts such as Anand, Bharuch, Vadodara, Dhar, Neemach, Jhabua, and Ratlam have higher LP (irrigated) (more than 2300 kg/ha) with the highest value for Ratlam (2573.96 kg/ha) than the other districts of the basin. The LP (irrigated) is lowest for the Aravali district of Rajasthan than the other districts. The Neemach district of Madhya Pradesh has the highest LP (rainfed) (1948.13 kg/ha) followed by Udaipur (1898.93 kg/ha) with the lowest value of LP for the Kheda district (1526.79 kg/ha) of Gujarat.
- The WP (irrigated) of all five districts of Madhya Pradesh is higher and vary from 1.46 kg/m^3 to 2.26 kg/m^3 than the districts of Rajasthan (varying from 0.79 kg/m^3 to 1.57 kg/m^3) and Gujarat (0.44 kg/$m^3$to 0.89 kg/m^3). The WP (rainfed) of all the districts of Gujarat is comparatively lower (varying from 0.34 kg/m^3 to 0.50 kg/m^3) than the districts of Madhya Pradesh (varying from 0.59 kg/m^3 to 0.70 kg/m^3) and the Rajasthan (varying from 0.48 kg/m^3 to 0.73 kg/m^3.
- As per the criteria are given by Safi et al. (2022) for classification of the districts as the 'bright spots-district' and 'hot spot-district, for irrigated cereals, the Ratlam district is found to have both high LP and WP as 2573.96 kg/ha and 2.14 kg/m^3, respectively among all the districts of the Mahi basin and hence the Ratlam district is classified as the 'Bright spot-district'. Similarly, the Anand district is found to have the lowest WP (0.44 kg/m^3) and the LP as 2467.51 kg/ha, and hence it is classified as the 'hot spot-district'.

Table A1
Input/Outputs required for WA+ Sheets.

Sheet No.	Sheet name	Inputs	Outputs
1	Resource-based sheet	Inputs from all the sheets	• Exploitable water • Available water • Utilized water • Reserved flows • Landscape ET • Storage change • Outflows • Utilizable outflow
2	Evapotranspiration sheet	▪ P ▪ ET ▪ LAI ▪ NDM (derived from NPP & GPP) ▪ PET	▪ E, T, I ▪ Beneficial ET ▪ Agricultural ET ▪ Managed ET
3	Agricultural services sheet	▪ ETGreen & ETBlue (derived from Budyko curve) ▪ ET ▪ P ▪ NDM ▪ LAI	▪ Land productivity ▪ Water Productivity
4	Utilized flow sheet	▪ P ▪ ET	▪ Withdrawals (SW and GW) ▪ Recoverable & Non-recoverable flows ▪ Consumed/ non-consumed water
5	Surface water sheet	▪ LAI ▪ ETGreen ▪ ETBlue	▪ Distinguishes fast and slow runoff ▪ Return flow from SW ▪ Inter-basin transfer
6	Groundwater sheet	▪ Surface runoff ▪ Incremental runoff ▪ Percolation ▪ Incremental Percolation ▪ Supply ▪ Storage change ▪ Soil moisture ▪ Saturated moisture content ▪ Baseflow ▪ Topographical map ▪ Flow direction ▪ EFR (environmental flow requirements) ▪ Water pollution level	▪ Recharge ▪ Withdrawals ▪ GW storage change ▪ Return flow to GW

Table A2
Information regarding Satellite-based datasets and sources.

Data	Temporal resolution	Spatial resolution	Availability	Data Source
Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)	Daily, Monthly	5 km	1981 onwards	https://data.chc.ucsb.edu/products/CHIRPS-2.0/
Actual Evapotranspiration (SSEBop)	Monthly	1 km	2003 onwards	https://earlywarning.usgs.gov/fews/datadownloads/Continental%20Africa/Monthly%20ET%20Anomaly
MOD-15; Leaf area index (LAI)	8-daily	500 m	2000 onwards	https://lpdaac.usgs.gov/products/mcd15a2hv006/
Gross primary production (GPP)	8-daily	500 m	2000 onwards	https://lpdaac.usgs.gov/products/mod17a2hv006/
Net primary production (NPP)	Yearly	500 m	2000 onwards	https://lpdaac.usgs.gov/products/mod17a3hv006/
IWMI LULC map	2000	250 m		http://waterdata:iwmi:org/applications/irri_area)
ESA-GlobCover LC v2	Annual	300 m	1992–2015	http://due.esrin.esa.int/page_globcover.php
MIRCA (Monthly Irrigated and Rainfed Crop Areas)	Annual	10 km	1998–2002	https://www.uni-frankfurt.de/45218023/MIRCA
WDPA (World Database on Protected Areas)	Annual	Shapefile	–	https://www:protectedplanet:net
JRC (Joint Research Center) Data Catalogue	Annual	30″	2017	https://data.europa.eu/euodp/en/data/dataset/jrc-floods-floodmapgl_permwb-tif
NRSC-250k	Annual	56 m	2004/05 –2015/16	https://bhuvan-app1.nrsc.gov.in/thematic/thematic/index.php

- As per the criteria are given by Safi et al. (2022) for classification of the districts as the 'bright spots-district' and 'hot spot-district, for rainfed cereals, the Neemuch district of Madhya Pradesh is classified as the 'Bright spot-district' with WP and LP as 0.59 kg/m^3 and 1948.13 kg /ha, respectively, and Anand as the hot spot- district with the lowest WP as 0.34 kg/m^3 and LP as 1572.21 kg/ha, respectively.
- Overall, at the state level (region level), the results show that the state of Madhya Pradesh is the 'Bright Spot-state' and Gujarat as the 'Hot spot-state' for irrigated and rainfed agriculture both.

7. Limitations of WA+ technique

Despite WA+ 's strong performance, one of the major limitations of WA+ technique is its reliance on satellite-based remote sensing datasets (due to the unavailability of the observed datasets such as ET, NPP, GPP, irrigated and rainfed crop maps, and others in many parts of the developing world e.g., India). The satellite data are prone to inaccuracy because they are used to estimate hydrological processes indirectly. Hence attention should be taken to enriching the satellite datasets

Table A3
Annual variation of Transpiration fraction (TF), Beneficial ET fraction (BF), Agricultural ET fraction (AGF), and Managed ET fraction (MF).

Year	TF	BF	AGF	MF
2003–04	0.47	0.47	0.61	0.66
2004–05	0.36	0.36	0.60	0.66
2005–06	0.38	0.38	0.61	0.66
2006–07	0.40	0.40	0.60	0.66
2007–08	0.46	0.45	0.61	0.66
2008–09	0.41	0.40	0.60	0.65
2009–10	0.36	0.36	0.61	0.66
2010–11	0.41	0.41	0.61	0.66
2011–12	0.38	0.38	0.61	0.66
2012–13	0.41	0.41	0.61	0.66
2013–14	0.48	0.47	0.61	0.66
2014–15	0.42	0.41	0.61	0.66
2015–16	0.40	0.40	0.60	0.65
2016–17	0.32	0.32	0.63	0.68
2017–18	0.39	0.39	0.61	0.66
2018–19	0.34	0.34	0.61	0.67
2019–20	0.44	0.43	0.70	0.75

Table A4
District-wise transpiration fraction.

Districts	Transpiration Fraction
Anand	0.42
Aravalli	0.39
Bharuch	0.27
Chhota Udaipur	0.31
Dahod	0.44
Kheda	0.38
Mahisagar	0.41
Panch Mahals	0.42
Sabar Kantha	0.42
Vadodara	0.40
Alirajpur	0.44
Dhar	0.39
Jhabua	0.43
Neemuch	0.46
Ratlam	0.40
Banswara	0.40
Chittaurgarh	0.40
Dungarpur	0.40
Pratapgarh	0.41
Udaipur	0.46

Table A5
Variation of Transpiration fraction on different land use classes.

Land use	TF
Irrigated rural areas	0.09
irrigated urban cities	0.06
irrigated crops-cereals	0.41
rainfed built-up area	0.25
fallow land	0.36
rainfed crops-cereals	0.45
moorland	0.43
natural lakes	0.17
grasslands	0.62
shrublands	0.49
closed evergreen forests	0.31
closed deciduous forests	0.45

through validation and ensemble processes (e.g., ET ensemble products by using satellite and observed ET datasets) during applications. WA+ estimates of WP and LP are highly dependent on the irrigated and rainfed crop map of the basin this necessitates for field surveys of cropping patterns for enhanced performance. Finally, the WA+ based LULC map (WALU) needs to be customized for local conditions (say region-specific) as the many classes (out of 80 classes) are not applicable to all the regions of the world. We took the cereals as the major crop class in the basin as without segregating them into wheat and paddy due to the absence of a detailed and accurate crop map area of the Mahi basin. This aspect also needs a major refinement to estimate crop-wise LP and WP in the basin. (Salvadore et al., 2020) also emphasized and accurate and detailed crop map of the basin to estimate the crop wise yield productivity. Heavy dependence of the WA+ technique on the satellite datasets limits this technique's applicability to future assessments incorporating climate change scenarios. The WA+ technique does not have the snow glacier component, and this limits its applicability to the glacier basins.

8. Future scope of research

In arid and semi-arid regions, agricultural activities have the largest proportion of the water consumption, and therefore this study was formulated to estimate the green and blue water consumption (ETBlue and ETGreen) and assess the spatial and temporal variability of the WP and LP to identify the bright-spots and hot-spots at the district administrative unit level for effectively managing the scarce water resources and sustaining food security in highly non-resilient basin of India. The findings will ultimately help formulate the "actionable management plans" to the stakeholders and policymakers of the Mahi basin. However, to address the limitations of the WA+ technique as discussed above, this technique should be coupled with a hydrological model say Soil and Water Assessment Tool (SWAT) and Spatial Processes in Hydrology (SPHY) model to address the current and future status of the water availability, demand, and supply scenarios. This ensemble modelling framework (WA+ coupled with the SWAT and SPHY) may also include the socio-economic aspects in the devising of actionable water management plans.

9. Recommendations/Suggestions

As the WP (both irrigated and rainfed) of the districts of Gujarat such as Anand, Vadodara, Kheda, and Bharuch is lower than that of the districts of Madhya Pradesh and Rajasthan in the Mahi basin and hence advanced irrigation techniques coupled with the crop diversification (adopting the technological and agronomical interventions of the bright spot districts such as Ratlam and Neemach) to increase the WP without effecting the LP of these hot-spot districts. The higher values of BWI are an indication of the more dependency on irrigation for agricultural production and hence efforts should be made to incentivize the farmers for crop diversification and shifting from more water intensive crops to less water-intensive crops. These findings can be taken as 'baseline' for developing 'actionable agricultural water management plans' by the stakeholders and policymakers of the basin states.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A

(See Appendix Figs. A1-A4 and Tables A1-A5).

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